Ninaivu

நினைவு · memory

Your family's photographs, at home.

The application guide · version 1.10.9

Ninaivu
the library

Mugil
backup

Sudar
the photo studio

Contents

Everything in this guide, and the page it is on.

Ninaivu 5

நினைவு · memory — your family's photographs, at home.

Ninaivu Lite 6
Where to start 7

01 Install 8

Ninaivu runs on a computer that stays on: a desktop under the stairs, a mini PC, an old laptop, a Mac mini, a Raspberry Pi.

Download 8
Windows 8
macOS 9
Linux and Raspberry Pi 9
Upgrading 10
What you see first 10

02 The first day 12

Right after the administrator is made, the console walks through the five things a new library needs.

1. Where are the photographs? 13
2. Who is in the household? 13
3. What should Ninaivu work out by itself? 13
4. A copy outside the house 13
5. On the household's phones 14
After that 14

03 Family and roles 15

Three roles, and every photograph has a visibility.

Visibility 16
Sharing 16
Faces and names 17
Uploads 17
Language 17
Dates 17

04 Backup 18

Three things can lose a library: the disk, the house, and a mistake.

Mugil — a copy outside the house 18
A second copy, and one off-site 20
Repairing a damaged file 20
Restore 20
The index 20
Phones 21
Is everything safe? 22
If something happens to you: the handover plan 22

05 Remote access 23

At home, Ninaivu answers to a name on the home network (ninaivu.local).

Home and the internet 24
HTTPS 24
The console from other devices 24
Everything Ninaivu sends out on its own 25

06 AI 26

Everything on this page runs on your own computer, and no photograph, face or name is sent anywhere by it.

Two kinds of computer 26
Adding search by description 27
What each pass does 27
Sudar 28
Extensions 29

07 Troubleshooting 30

The phone cannot open Ninaivu 30
I cannot open the console from my laptop 30
A profile will not open away from home 30
Someone forgot their password or PIN 30
The browser warns about the certificate 31
Nothing appears after choosing the folder 31
The scan is slow 31
An import says some files could not be copied 32
Uploads say there is not enough free space 32
A model would not download 32
Mugil says the copy did not restore 32
Mugil says nothing is uploading 33
Ninaivu will not start 33
Moving to another computer 33
Getting help 33

A The family app 34

Everything a family member or guest needs, and nothing else.

Coming in 35
The gallery 36
A photograph 37
People 39
Albums 40
Photo books 40
Sending photographs 40
Scanning old prints 41
The Photo Studio 41
Sudar, the photo studio 43
Language 45
The screen lock 45
Sound recordings 45
On a phone 46

B The console 47

Everything the administrator does: getting a library indexed, deciding who sees what, and the tools — Import, Faces, Mugil, Health.

Home — Overview 48
Library 49
Review 52
People & access 54
Backup & health 57
AI 59
System 60
The command line 62

How to read this guide

Note
Worth knowing

Tip
A quicker way

Warning
Read before you go on

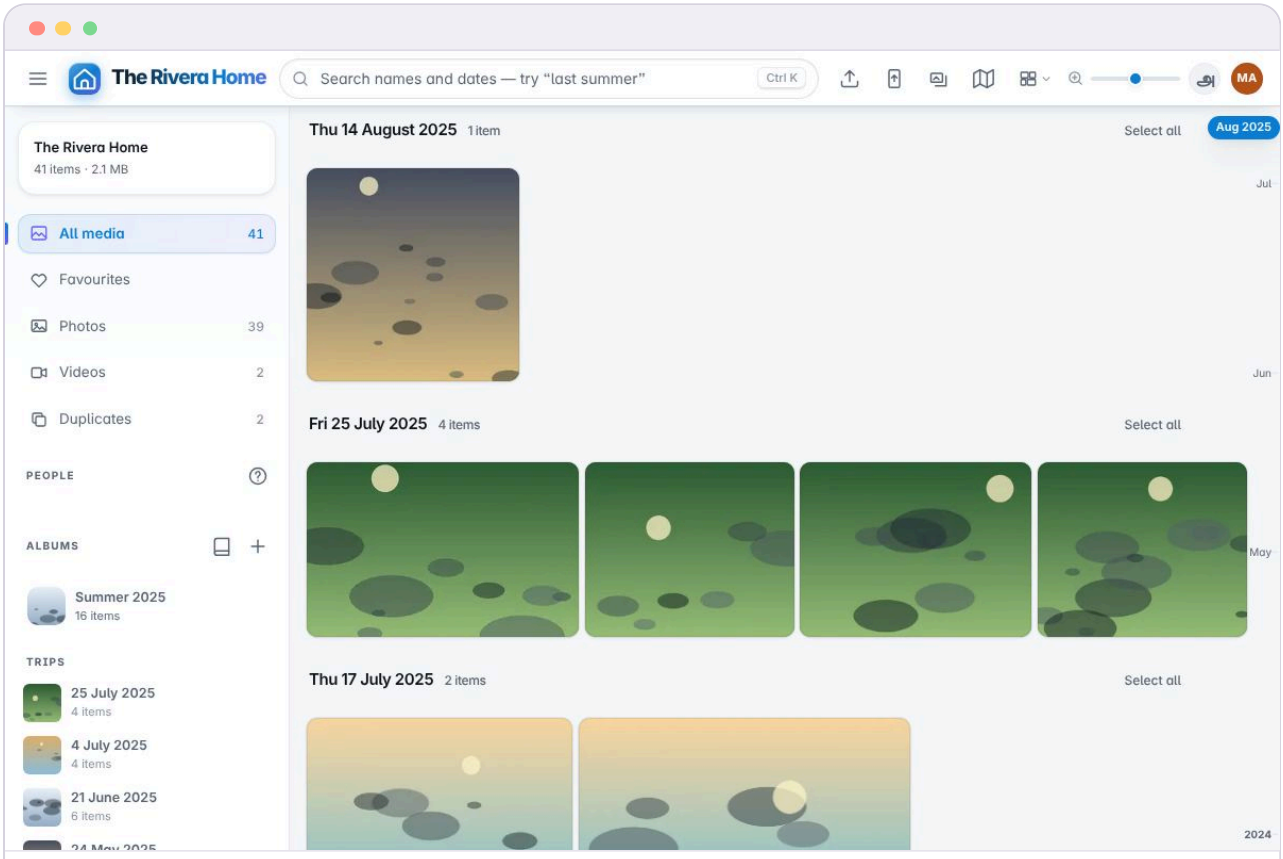
The pictures use a generated sample library, not anybody's photographs. Everything here runs on a computer you own; nothing leaves the house unless you choose where it goes.

NINAIVU 1.10.9

Ninaivu

நினைவு · *memory* — your family's photographs, at home.

Ninaivu is a private photo and video library for one household. It runs on a computer you own, indexes what is on your own disks, and gives your family what people usually leave to a cloud service: faces and names, search by description, places on a map, "on this day", a link for the grandparents, and phones that send their photographs home. Nothing leaves the house unless you choose where it goes — and the backup goes encrypted.



The gallery

It has three parts, each with a name:

Ninaivu	The library: gallery, faces, search, places, albums, sharing, and the three roles
Mugil (**முகில்**, ***cloud***)	Backup: an encrypted copy of the library on Google Drive, and the household's phones sending their photographs to the house
Sudar (**சுடர்**, ***glow***)	The photo studio: light, colour, straightening, crops and suggestions, in the browser, never changing the original

Ninaivu Lite

Ninaivu Lite `github.com/javajaga-usa/Ninaivu-lite` is the smaller edition for a household that wants a gallery on the home Wi-Fi, with fewer dependencies. It uses Ninaivu's family gallery, admin console and share screens, in Tamil and English, and reads your photo folders without changing the originals.

CHOOSE NINAIVU LITE FOR	CHOOSE NINAIVU FOR
Timeline, folders, favourites, albums and search by name, folder, date or camera	Face recognition, search by description and places on a map
Admin, Family and Guest roles, visibility controls and password-protected share links	The full library with Mugil cloud backup and phone backup
Importing old drives into a hash-checked archive and editing with Sudar in the browser	Model-backed AI tools and the full Sudar studio
A lighter engine built on Flask, Pillow and waitress	A computer with capacity for the larger engine and optional models

Lite has no model-backed AI, cloud backup or phone backup. It runs on Windows 10+, macOS, Linux and Raspberry Pi 4/5 with a 64-bit OS. Installers carry their own Python; a portable Windows zip and Docker setup are available too.

- **Download Lite** `github.com/javajaga-usa/Ninaivu-lite/releases/latest` — choose the installer for your computer. Releases are currently unsigned; see the installation notes `github.com/javajaga-usa/Ninaivu-lite#install`.
- **Lite user guide** `github.com/javajaga-usa/Ninaivu-lite/blob/main/docs/USER-GUIDE.md` · **தமிழ் வழிகாட்டி** `github.com/javajaga-usa/Ninaivu-lite/blob/main/docs/USER-GUIDE.ta.md`.
- **Moving from Lite to Ninaivu** `github.com/javajaga-usa/Ninaivu-lite/blob/main/docs/UPGRADE.md` — an optional move if you need the larger edition later.

Lite uses HTTP and is intended for the home network. Keep it on your home Wi-Fi; do not expose it directly to the internet. The rest of this documentation describes the full Ninaivu edition; use the Lite guides above for Lite setup, backup and recovery.

Where to start

1. Install p. 8 — the installer for Windows, macOS, Linux or a Raspberry Pi. Ten minutes.
2. The first day p. 12 — the folder, the household, what the scan works out by itself.
3. Family and roles p. 15 — who sees what.
4. Backup p. 18 — a copy outside the house, tested every week.
5. Remote access p. 23 — reaching it from away.
6. AI p. 26 — what runs on your machine, and what a small machine skips.
7. Troubleshooting p. 30.

Then **the guide**, everything each screen does: the family app p. 34 for the household, the console p. 47 for whoever runs it. Every screen, pictured: a tour of the screens.

> **Status**
>
> Ninaivu 1.10.9 is the general edition of a server that has run one household's library for a year. It has over 4,000 tests. The roadmap says what comes next.

CHAPTER 1

Install

01

Ninaivu runs on a computer that stays on: a desktop under the stairs, a mini PC, an old laptop, a Mac mini, a Raspberry Pi. Download the installer for that computer and run it. It brings everything Ninaivu needs with it, so nothing else has to be installed first.

Download

The installers are on the Ninaivu download page, github.com/javajaga-usa/Ninaivu/releases/latest. Pick the file for your computer:

COMPUTER	FILE
Windows	`Ninaivu-<version>-windows-x64.exe`
Mac with Apple silicon (M1 and later)	`Ninaivu-<version>-macos-arm64.dmg`
Mac with an Intel processor	`Ninaivu-<version>-macos-x86_64.dmg`
Linux PC	`Ninaivu-<version>-linux-amd64.sh`
Raspberry Pi 4 or 5 (64-bit)	`Ninaivu-<version>-linux-arm64.sh`

The same page has this guide as a PDF, in English and Tamil.

Windows

Run `Ninaivu-<version>-windows-x64.exe`. It installs for your account only (no administrator prompt), puts a private Python and everything Ninaivu needs under `%LOCALAPPDATA%\Programs\Ninaivu`, and adds **Ninaivu** to the Start menu. Leave *Start Ninaivu at sign-in* ticked and it is always there. **Ninaivu** in the Start menu, and **Ninaivu Control Panel** on the

Desktop, open the Control Panel: Start, Stop and Restart, the addresses to open, the computer's readings and the log. *Ninaivu tray*, in the same Start menu folder, is the small menu in the system tray for anyone who prefers that.

If the installer is not signed (the release notes say), Windows SmartScreen says *Windows protected your PC*: choose **More info → Run anyway**.

More on the Control Panel and the tray.

macOS

Open the `.dmg` for your Mac — `arm64` for Apple silicon, `x86_64` for Intel — and drag **Ninaivu** to Applications.

If the app is not notarised (the release notes say), macOS will not open it the first time. Try once, then go to **System Settings → Privacy & Security** and choose **Open Anyway**; after that it opens normally.

Opening **Ninaivu** opens the Control Panel: Start, Stop and Restart, the addresses to open, the computer's readings and the log, and *Start Ninaivu when I sign in*. The menu-bar tray is there too, for anyone who prefers it.

Linux and Raspberry Pi

Use `Ninaivu-<version>-linux-amd64.sh` for a PC, or `Ninaivu-<version>-linux-arm64.sh` for a Raspberry Pi 4 or 5 running a 64-bit OS (or any other 64-bit Arm board). Open a terminal in the folder it was downloaded to and run it:

```
sh Ninaivu-<version>-linux-arm64.sh
```

No root is needed, and the machine needs no Python: the file carries its own, with every package Ninaivu uses, and downloads nothing. It asks once where the photographs are, installs Ninaivu for your account, puts **Ninaivu Control Panel** in the applications menu and on the Desktop, sets Ninaivu to start when the computer starts and keep running after you sign out, starts it, and prints the address to open.

Upgrading

Ninaivu does not look on the internet for newer versions. When you want one, download the newer installer from the same page.

Stop Ninaivu first. In the Control Panel press **Stop** and wait until it says Stopped, then close the Control Panel. Then install the newer version the same way as the first: run the new Windows installer, drag the new Ninaivu into Applications on a Mac, or run the new `.sh` file on Linux. It replaces the program, and only the program. The Windows and Linux installers also ask a Ninaivu that is still running to stop properly; on a Mac, always stop it yourself before dragging the new one over the old.

Your photographs are never touched by an update. The settings, the index (names, albums, faces, share links), the people's accounts and the AI models you downloaded are kept as they are too.

The first time the new version starts, it makes a copy of the settings and the index before it changes anything, then starts as before. If you ever need to go back to the older version, that copy is how the technical administrator puts things back as they were (see the technical administrator guide, "Updating to a new version").

> **Other ways to run Ninaivu**
>
> Docker on a NAS, running it from the source code, and the command-line options are for whoever looks after the technical side. They are in the technical administrator guide, in English.

What you see first

The first screen makes the administrator. On the computer Ninaivu runs on, that is all, whether you open it as `localhost`, as `ninaivu.local` or by the computer's own address. From any other device it also asks for the **setup code**, so nobody else at home can claim a new library first. The code is printed where Ninaivu started (the Control Panel's log, or the last lines in the terminal) and kept in `setup-code.txt` in Ninaivu's state folder (the lines that print the code show its full path) until the administrator exists; it stays the same across restarts. After that the console walks you through the first day p. 12.

Two addresses

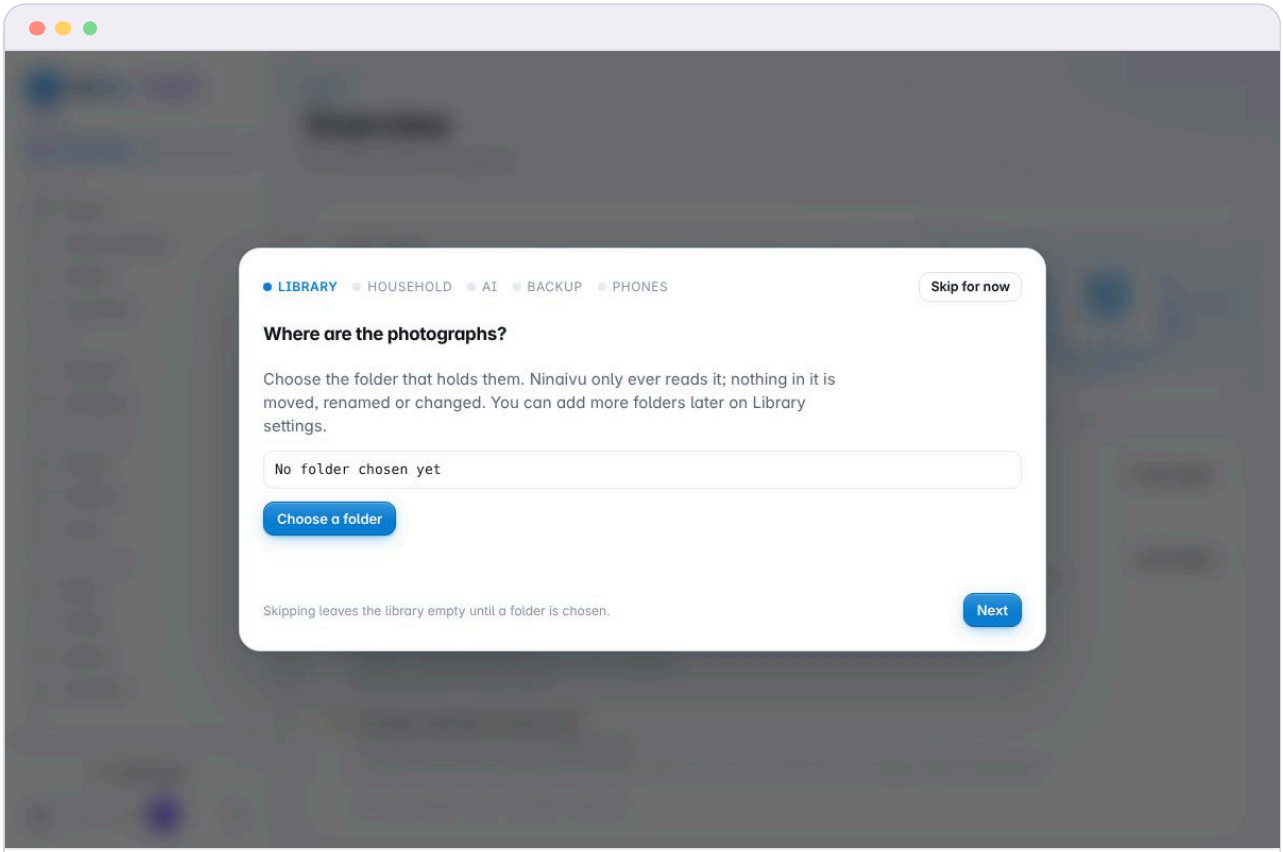
Ninaivu has two faces on two ports: the **family app**, where everyone looks at the library, and the **console**, where the administrator runs it. The console answers on this computer only until you open it to the home network on the Server page; the family app is on the home network.

CHAPTER 2

The first day

02

Right after the administrator is made, the console walks through the five things a new library needs. Each step can be skipped and each is just a page of the console, so nothing here is a second way of doing anything.



The first day

1. Where are the photographs?

Choose the folder that holds them. Ninaivu only ever reads it: nothing is moved, renamed or changed. You can add more folders later on **Library settings**, and sweep old drives and memory cards into one dated archive with **Import** — which also understands a Google Photos Takeout export, dates, places, descriptions and albums included.

2. Who is in the household?

A **family member** taps their name and enters a PIN. A **guest** signs in with a password (or a PIN, or a tap, set on **People** later) and sees only what you make public. Add the people you know of now; the rest can come later on **People**. A profile without a PIN can be opened from any phone or computer on your home network, so give a PIN to anyone whose photographs should stay theirs. More on roles. p. 15

3. What should Ninaivu work out by itself?

Everything here runs on this computer; no photograph leaves it. **Find the people in photographs** needs a 37 MB model, downloaded when you turn it on. **Name the places** needs an 11 MB list of places, downloaded once from GeoNames. **Read the words in photographs** needs the text reader from Extras. **Search by description** — finding "the beach at sunset" without anybody tagging it — is about 2 GB, added only if you press *Add search by description*; it is worth it on a computer with 8 GB of memory or a graphics card. Each pass takes time on a large library and can be turned off part-way. More on AI. p. 26

4. A copy outside the house

A fire, a theft or a failing disk would take every photograph. **Mugil** keeps an encrypted copy on Google Drive and checks once a week that it restores. It needs a Google account and an encryption key — nothing is uploaded until the key is made — which take a few minutes on the Mugil page; the Overview reminds you until it is done. More on backup. p. 18

5. On the household's phones

Open the address shown on a phone on the home Wi-Fi and add it to the home screen: it opens like an app. From its backup screen, the photographs and videos you choose are sent to the house; keep the page open while it sends.

After that

The console's **Overview** answers two questions before anything else: *is everything safe?* and *what needs you?* — uploads to approve, faces without a name, photographs that may be on their side. Come back to it now and then.

CHAPTER 3

Family and roles

03

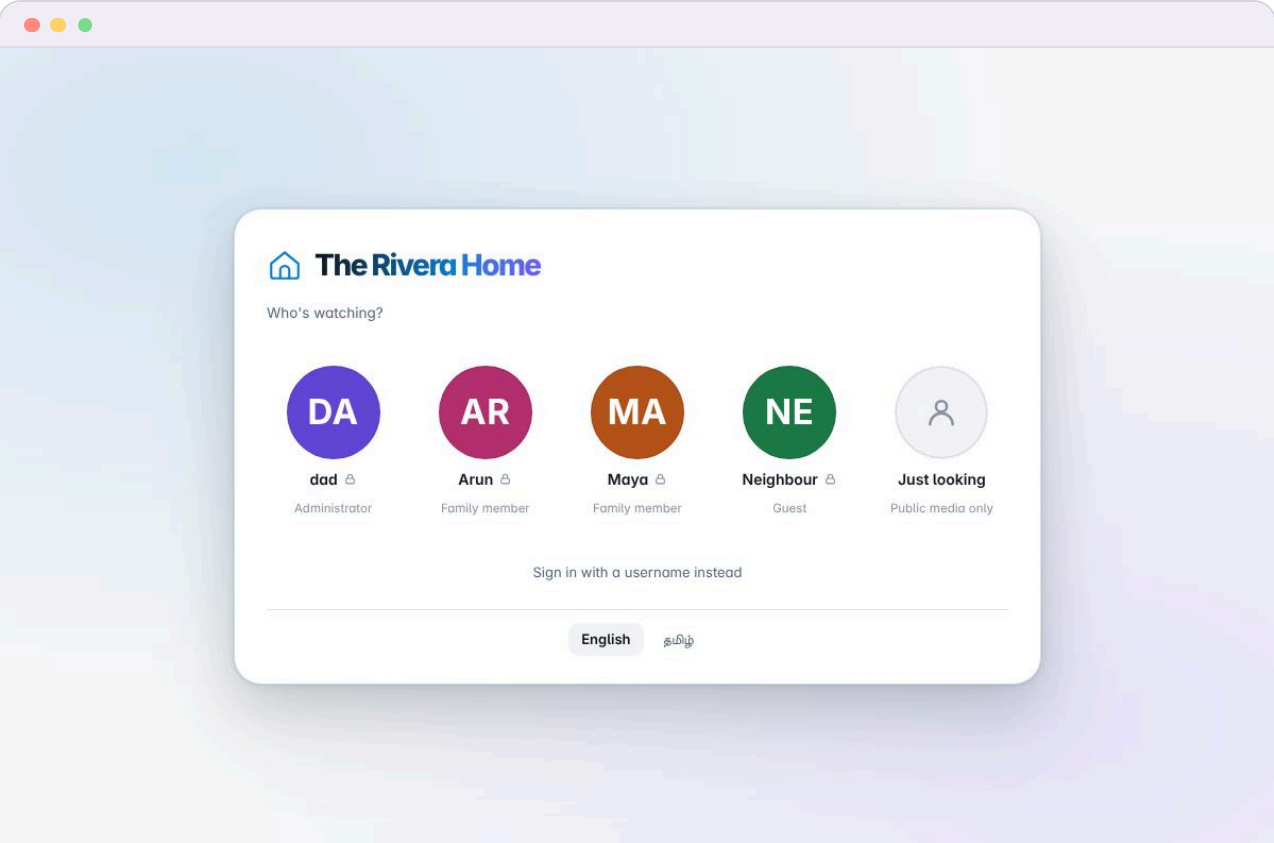
Three roles, and every photograph has a visibility. That model is the product; everything else is built on it.

ROLE	SIGNS IN WITH	SEES
Administrator	a password, on the console (and in the family app, from their locked tile on the profile picker, where the same account can also delete and change visibility)	everything, and runs the server
Family member	a PIN, a password, or a tap, from the profile picker	everything marked for the family
Guest	a password, a PIN, or a tap	only what was chosen for them

A profile set to *tap to enter* needs no secret, which suits a shared tablet — and means any device on the home network can open it. Give a PIN to anyone whose photographs should stay theirs. From the internet such a profile never opens, and the library cannot be browsed without signing in: only the home network, Tailscale and WireGuard count as home (remote access p. 24).

With **open browsing** on, a visitor can look at what is public without signing in at all; off, the whole library is private. Screenshots, documents and photographs of screens are admins-only by default (*hide screens*), and so is every sound recording. Nothing admins-only is read by AI: it is not tagged, the text in it is not read, faces are not looked for in it, and it is never sent to Gemini. Whatever was read before it was hidden is removed when it is hidden.

Both apps lock after 15 minutes without use; the person's PIN or password opens them again, and whatever was running carries on meanwhile.



The profile picker

Visibility

Every folder has a setting — everyone, family, or admins only — and every photograph can be set on its own. **People & access → Visibility** holds the rules for everyone first (whether visitors may browse, whether explicit content is screened, how far back in time each role may look) and then each folder's own setting. Every change can be undone in one step.

A family member can also be given a folder of their own: a private corner of the library that only they and the administrator see.

Sharing

Family members and administrators can make a link for one photograph or an album, with an optional password, scoped to what its maker could already see. Guests cannot share.

Faces and names

People & access → Faces shows the people the face matcher found, grouped. Name one face and the rest of its group follow. Faces are found and matched on this computer. The names are part of the index, so they travel only where the index does: into Mugil's daily copy, encrypted with your key.

Uploads

Photographs the family send from their phones wait under **Review → Uploads** for the administrator to file or refuse. A family member's edit in Sudar waits the same way. Nothing a family member does changes the library without the administrator seeing it.

Language

Ninaivu is in English and Tamil. Each person chooses on the language button (the sign-in and lock screens have it too), and the choice is saved on their profile, so it follows them to every device. In Tamil, the name is நினைவு.

Dates

A role can be limited to a window of time — guests see this year's photographs, say, and nothing older. Date access.

CHAPTER 4

Backup

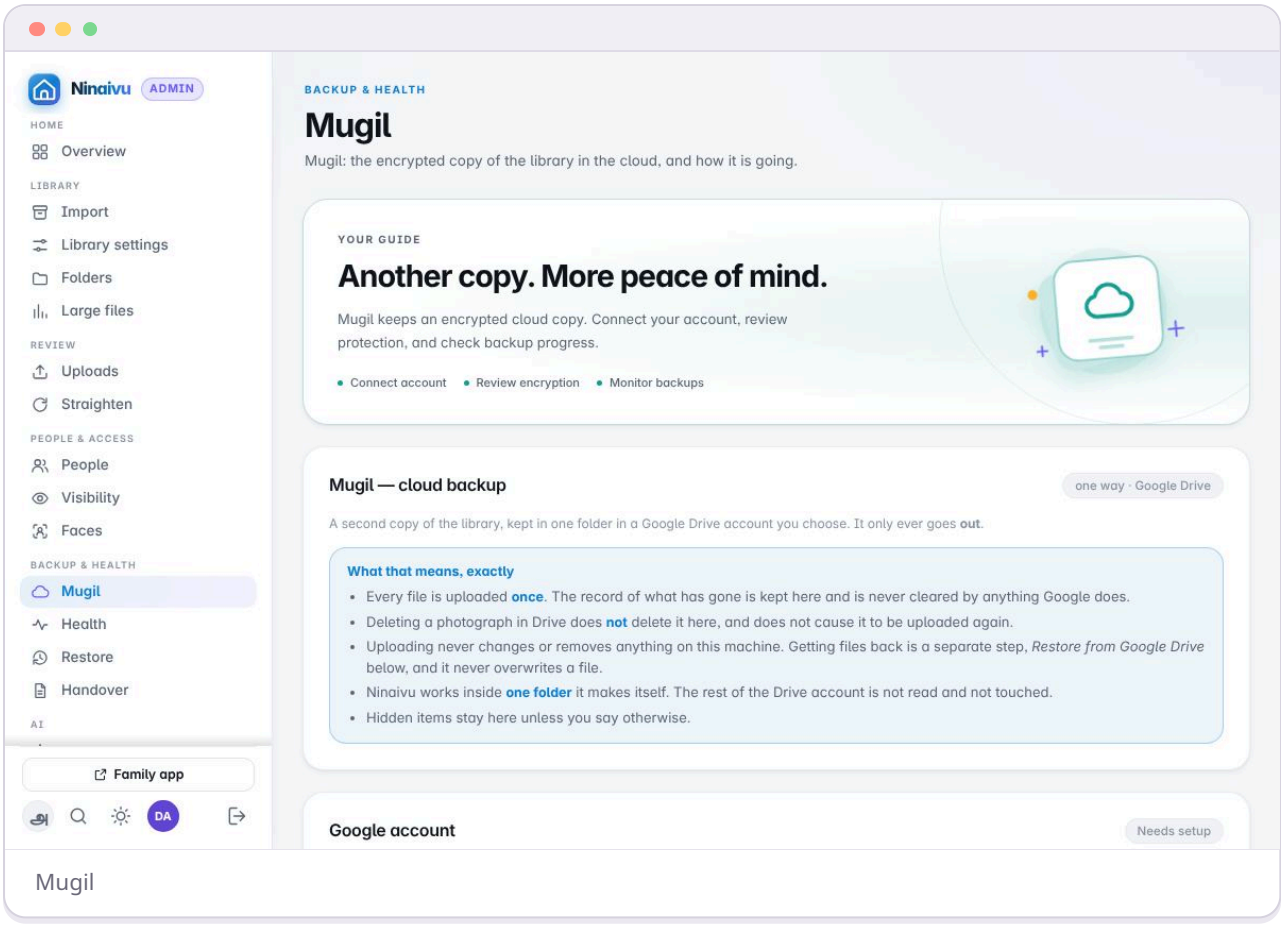
04

Three things can lose a library: the disk, the house, and a mistake. Ninaivu keeps a copy against each.

Mugil — a copy outside the house

Mugil (முகில், *cloud*) keeps an encrypted copy of the library on Google Drive. It is one way: nothing in Drive can change, move or remove anything at home, and deleting a photograph from Drive does not delete it here.

Encryption is on from the start, and **nothing is uploaded until you have made the encryption key** on the Mugil page — so no photograph, and no copy of the index with the names given to faces, ever goes up unencrypted. An administrator can switch encryption off; the page then says plainly that photographs and names go up as they are.



Mugil

On the **Mugil** page: the Google account (a few minutes, once), the encryption key and its **recovery file**, what is sent and what is kept back, the speed and the hours it may use, and **Test restores** — the weekly test that brings a few files back and checks they match, with its history.

> ⚠ **The recovery file**
>
> The copy in Drive is useless without the key. Ninaivu writes a recovery file and asks you to keep it somewhere that is not this computer — a USB stick in a drawer, a password manager. Without it, a lost computer means a lost library, however good the copy in Drive is. What is in it.

Google Drive is Mugil's destination today; the second disk and S3 are the separate copies below, and WebDAV is on the roadmap.

What is sent can be narrowed in **Settings → Backup**: everything, all but videos, or pictures only; nothing over a size; and folders or words in a path to leave out. A file over **cloud approval** (1 GB by default) waits until an administrator says yes to it, and the safety check says when files are waiting.

A second copy, and one off-site

Two more copies, set up beside Mugil:

- **A second copy** on another disk or a NAS share: every photograph copied across on a schedule and checked against its own record every few weeks. Nothing removed at home is removed there.
- **An off-site copy** to any S3-compatible bucket (Backblaze B2, Wasabi, MinIO, Amazon S3) or a folder (a friend's disk, a mounted share), encrypted with the Mugil key before it leaves. It can be put back on any computer, with or without Ninaivu running, by `ninaivu offsite-restore --recovery <recovery file> <copy> <output>`.

The safety check **Every photograph in more than one place** counts the photographs that have no copy anywhere but this computer.

Repairing a damaged file

The scheduled check (every 30 days by default) reads every photograph and compares it with what was recorded when it was indexed. With **scrub repair** on, a file that has changed on disk without being edited is put back from the second copy or from Mugil, and the damaged one moved out of the library into the state folder (`repair/damaged/`), never thrown away.

Restore

Backup & health → Restore brings photographs back from the cloud copy in three steps: what to bring back, where to put it, and the key. Nothing here ever overwrites a file that is already there.

The index

The index — faces and the names given to them, albums, who may see what — is copied on this computer once a day, seven copies kept (**Health** shows them), and into Drive with the library, encrypted with the same key, so a lost computer loses nothing but time. To put a copy back: stop Ninaivu and run `ninaivu restore <file>`. Moving to another machine.

Voice stories (the recordings people make about a photograph) live in the `stories` folder beside the index, not in the library. The daily copies on this computer carry them, words and sound, and a restore of one puts them back. The copy of the index in Drive leaves them out on purpose —

nothing about a story is sent to any outside service — and so do the second disk and the off-site copy, which hold the photographs only. To keep the stories somewhere else too, keep the daily copies' folder somewhere else (see **Backup & health**).

Phones

On a phone, the family app's backup screen sends the photographs and videos you choose to the house over the home Wi-Fi. A web page cannot read the phone's library by itself or keep going in the background, so you pick what to send and keep the page open while it sends; it skips what is already here and carries on where it stopped. What arrives waits under **Review → Uploads** for the administrator, unless the administrator switched on **trust phone backups**; an administrator's own phone skips the review.

Backing up by itself

A sync app on the phone (PhotoSync, on iPhone and Android, is one) can send new photographs home without anybody opening a page. Each phone gets its own **phone key**, made by its owner in the family app:

1. On the phone, open **Back up this phone**, check the phone's name, and under **Back up automatically** press **Make a key for this phone**. Copy the key; it is shown only once.
2. In the sync app, add a **WebDAV** destination: the address Ninaivu gives with the key (it ends in `/dav/`), your username, and the key as the password.
3. Turn on the app's automatic transfer, for example whenever the phone arrives home or joins the home Wi-Fi.

What the app sends goes through the same checks as the backup screen: it is recognised if the library already has it, waits under **Review → Uploads** unless phone backups are trusted, and is filed by the date it was taken whatever folder the app chose. A key can only add photographs: it cannot open, list or delete anything in the library. The inbox answers only at home and over Tailscale or WireGuard, never the internet. Revoke a lost phone's key under **Back up automatically** in the family app, or anyone's from the console under **Review → Uploads → Phone keys**, which also shows what each key has sent and when.

An iPhone lets such an app send only when its trigger fires, so expect new photographs to arrive when the phone gets home rather than the moment they are taken. One file can be up to **phone_upload_max_gb** (8 GB unless changed); the backup screen has no such limit.

Is everything safe?

The console's Overview runs the household's safety checks every time it opens — a copy outside the house, a backup that restores, copies of the index, healthy drives — and says what to do about each.

If something happens to you: the handover plan

Usually one person knows how the library works. **Backup & health → Handover** writes down who takes over and how, so the photographs outlive the person who looks after them.

- Name the **successors**, in order, from the family members' profiles, and write them a message: where the backups are, where the backup key's recovery file and the computer's password are *kept* ("the blue folder in the steel almirah"), how to reach the computer, who can help. Never type a password or the key itself; text that looks like the key is refused.
- Ninaivu fills in the rest and keeps it current: the computer's name, its version, the library folders and its own folder, every backup with when it last worked, the backup key's fingerprint (never the key), how the house reaches it from outside, and how to bring the photographs back.
- **Print the handover sheet**, in English or Tamil, and keep it with the family's papers. Print it again after a change. The Overview reminds you when there is no plan, when it has not been looked at for six months, or when the backups changed since.
- **Make a code**: a one-time handover code, shown once and kept only as a hash, printed on the sheet or written on it by hand. Making a new one ends the old one.

A successor signs in to their own profile in the family app, opens it, and under **Take over as administrator** types the code and their own password (or, for a profile that opens with a PIN, the PIN, and then chooses the password they will use as administrator). A profile that opens with a tap has nothing of its own to check: give it a PIN or a password first. Only somebody named in the plan can use it, and guesses are limited. Nothing happens at once: for the days the plan says (7 unless you choose 0 to 30), every administrator sees it at the top of every page and can **stop this handover**, which uses the code up. When the wait is over the successor becomes an administrator; nobody else's role changes. Every step is in **System → Activity**.

CHAPTER 5

Remote access

05

At home, Ninaivu answers to a name on the home network (`ninaivu.local`). Away from home, it is reached through something the household set up — and Ninaivu treats *how* as a choice, made once on **System → Server → Away from home**.

CHOICE	WHAT IT IS	WHAT IT NEEDS
Tailscale	A private network between your own devices; the easiest	Tailscale installed on the server and on each phone
WireGuard	The same idea, set up by hand	the tunnel, and its subnets typed here
A tunnel	Cloudflare Tunnel or the like: a public name that reaches the house	the tunnel, and the name typed here
Your own reverse proxy	Caddy or nginx in front, with a certificate	the proxy (examples)
Nothing	Home network only	—

The default works out Tailscale by itself when it is on the machine.

Each choice says what it still needs, decides which addresses count as *away from home*, which addresses are listed for the household, and whose certificate is served.

Home and the internet

Whatever is chosen above, Ninaivu decides for every request whether it came from home or from the internet. **Home** is the house's own network (IPv6 included), the computer itself, Tailscale and WireGuard. **The internet** is any public address, which is what a port forwarded on the router brings in, a tunnel or proxy Ninaivu was not told about, and Tailscale Funnel. From the internet:

- a profile set to *tap to enter* does not open, and the library cannot be browsed without signing in; give a PIN to anyone who uses Ninaivu through a tunnel;
- plain HTTP is refused, because passwords, PINs and share links would cross the internet readable: reach Ninaivu with Tailscale, or put HTTPS in front of it;
- the console does not open at all (below);
- wrong administrator passwords pause sign-in by username for longer each time: after 20 in half an hour, for 30 minutes, then an hour, then two. The computer Ninaivu runs on can always sign in.

The safest setup is still the simplest: Tailscale for away from home, and no port forwarded on the router (switch UPnP off on the router too). Then nothing on the internet can connect at all.

Running the Docker image, Ninaivu cannot see where a connection came from: Docker hands every one over from its own address. Everyone is then treated as the internet — profiles need a PIN, and nobody browses without signing in — until a reverse proxy in front passes the real address on (`trusted_proxies`, in the technical administrator guide).

HTTPS

Ninaivu serves plain HTTP unless HTTPS is turned on: the **HTTPS** switch on the console's Server page, which restarts Ninaivu. An installation that already served HTTPS before 1.10.9 keeps it. With HTTPS on, Ninaivu makes its own certificate authority, and each device that should not see a warning trusts that certificate once; the tray and the Server page have the button (shown only while Ninaivu is on HTTPS), and the tray page has the details for Windows and macOS. Tailscale and a reverse proxy bring their own certificates.

The console from other devices

The family app answers on the whole home network; the **console answers on this computer only** until you tick *Open this console from other devices at home too* on the Server page (Ninaivu restarts). `--admin-host` sets its address outright. Opened to the network, it is still for home, Tailscale and

WireGuard only, and refuses the internet; `console_from_internet` (**Advanced settings → Remote access**) opens it to a tunnel or proxy on purpose, which is seldom wise.

Everything Ninaivu sends out on its own

Remote access is a way *in* to the house. Faces, names and photographs are never sent anywhere by it. This is the complete list of what Ninaivu asks of the internet without a person pressing something at that moment:

WHAT	WHEN	HOW TO STOP IT
Mugil: the photographs and the daily copy of the index, encrypted, to the Google account you connected	While Mugil is on	**Mugil**
Tailscale's HTTPS certificate for this computer, through Tailscale	With HTTPS on and Tailscale installed, every few weeks. The name `<computer>.<tailnet>.ts.net` then appears in public certificate logs, as every Let's Encrypt name does.	**Advanced settings** → `tailnet_https`
The image model's weights, from Hugging Face	Once, at the first start after search by description was added; nothing after that	Do not add search by description
A model download that stopped half-way	At the next start, until it finishes	**AI models**
The list of places, from GeoNames	Once, at the first scan after *Name the places* was switched on	Leave it off
Map tiles, from OpenStreetMap — they show roughly where your photographs were taken	Whenever someone opens the map, if `map_tiles` is on (off by default; the map uses a built-in outline otherwise)	**Advanced settings** → `map_tiles`

Model downloads, notifications by email or webhook, and extensions such as Gemini happen only once an administrator has set them up.

CHAPTER 6

AI

06

Everything on this page runs on your own computer, and no photograph, face or name is sent anywhere by it. The one exception is an extension you switch on that says otherwise on its switch (Gemini, below). The models themselves are downloaded once, when you turn their feature on.

Two kinds of computer

Ninaivu is tested on two kinds of machine, and works out which yours is when it starts (**System → Performance** says which, and why):

	BASIC	FULL
The machine	anything else: a mini PC, an old laptop, a single-board computer, a desktop without a graphics card or without the image model	a graphics processor with at least 6 GB of memory, or Apple silicon, or any computer with 8 GB of memory once search by description is added
The gallery, albums, sharing, roles	yes	yes
Faces, grouped and named	yes	yes
Sideways photographs put right	yes	yes
Places named from the location	yes	yes
Text read from photographs (with the text reader)	yes, slowly	yes

	BASIC	FULL
Search by words in names, dates, places and people	yes	yes
Mugil, the encrypted copy	yes	yes
Tags, "the beach at sunset" search and descriptions from the image model	—	yes
Finding visually similar photographs (`S` in the viewer)	—	yes
Videos described from several moments	—	yes
Creative Studio: generative edits, object removal, upscaling	—	with a graphics card

On a Basic machine, *auto* means the light engine, so it never tries to load the large models and starts quickly. Choosing the image model outright (`--ai clip` , or `ai_engine` on **Advanced settings**) still loads it. Setting `hardware_tier` on **Advanced settings** overrides the measurement.

Adding search by description

It is not installed by default: it is about 2 GB, plus the model itself. Add it from the first-day walk-through or **Home & extensions → Extras → Search by description**, then restart Ninaivu. On a Full machine it is used from then on; the model's weights are fetched once at that first start, and after that Ninaivu tells the model libraries to make no request at all (`HF_HUB_OFFLINE`).

What each pass does

AI → AI models has a switch for each, saying which model it needs and whether that model is here.

- **Faces** — a 37 MB detector and matcher. Groups are named on **Faces**; name one and the rest follow.

- **Places** — an 11 MB list of places, downloaded once from GeoNames; the photograph's own location becomes "Paris" or "the beach". The map uses a built-in outline of the world; street-map tiles from OpenStreetMap are off unless `map_tiles` is switched on, because each tile asked for shows roughly where a photograph was taken.
- **Sideways photographs** — a 77 MB model that says which way up a photograph goes. By default it looks by itself after every scan and turns what it finds straight away, as one batch that **Undo** on **Review → Straighten** puts back; switch that off and what it finds waits there for approval.
- **Text in photographs** — the text reader from **Extras**; slow on a processor.
- **The image model** (Full, once added) — CLIP or SigLIP: tags, "the beach at sunset" search, descriptions, visually similar photographs.
- **Videos** (Full) — described from several moments rather than the poster frame.

Sudar

Sudar (சுடர், *glow*) edits in the browser: light, colour, straightening, crops, looks, and suggestions measured from the picture. "Make it warmer" in plain words. The original is never changed; a copy is saved beside it, and so can any result a tool makes — a cut-out, a colourised print — a family member's copy waiting for the administrator's approval like an upload. The **Photo Studio** (*Edit photo* in the viewer) renders through the same engine, so a slider means the same thing in both; the family guide p. 41 has every panel.

Its **Skin** and **Hair** tools work on every person in the photograph, each judged against their own skin, using the face model from **AI models**. They leave a bindi, sindoor or sacred ash exactly as it is, and have no setting that lightens a complexion. Hair the colour of what is behind it is found with the help of *Background removal (RMBG-1.4)* when that model is installed, and painted in by hand otherwise; **Add hair** draws new hair from the strands beside it, on this machine. See the family guide p. 44.

Its **Enhance tools** each need a model, and the card says which:

- **Upscale** — *Upscale ×4 (Real-ESRGAN)*, from **AI models**.
- **Restore faces** — *Restore faces (GFPGAN 1.4)*, from **AI models**.
- **Colourise** — a DDColor ONNX model. The catalogue cannot pin one yet, so this is the one model placed by hand: put `ddcolor.onnx` in the `onnx` subfolder of the models folder (**AI models** shows the folder), or point `NINAIVU_COLORIZE_MODEL` at the file. With Restore faces too, **Revive old photo** does both in one tap.

Each runs on this machine and nothing leaves it; with the Creative Studio extension, an AI server workflow can take any of them instead.

Extensions

Anything that cannot keep the core's two promises — nothing leaves the house unless you chose where, and it runs on a small machine — is an extension, off until switched on under **System → Home & extensions → Extensions** (it takes effect when Ninaivu next starts), and its switch says what it does:

- **Creative Studio** — generative edits and the heavy tools with large models, on this computer with a GPU or on a ComfyUI server at home. The AI server page refuses an address on the internet, so nothing leaves the house (a Tailscale address counts as home).
- **Gemini** — edits and descriptions from Google's models. A re-encoded copy of the photograph goes to Google when, and only when, a request names it. Only an administrator can send one, unless *Let family members send photographs to extensions that go outside the house* is ticked under Home & extensions → Extensions.

Setting up extensions is covered in the technical administrator guide.

CHAPTER 7

Troubleshooting

07

The phone cannot open Ninaivu

- The phone must be on the home Wi-Fi, not mobile data.
- Use the address the console's **Server** page lists under *for the household*. `localhost` only works on the server itself.
- **Network access** on the Server page must be on; off, Ninaivu answers on the server only. Turning it on needs a restart.
- Windows: if the firewall blocks it, the startup log prints the two `netsh` commands that open the ports; run them once in a terminal opened as administrator.

I cannot open the console from my laptop

The console answers on the computer Ninaivu runs on only, until you tick *Open this console from other devices at home too* on the Server page (at that computer). Ninaivu restarts, and the console is then on the home network too. Away from home, open it over Tailscale or WireGuard: it does not answer the internet, so it will not open through a forwarded port or a tunnel.

A profile will not open away from home

Through a forwarded port, a tunnel or Tailscale Funnel, Ninaivu is on the internet, and a profile without a PIN does not open there; plain HTTP is refused altogether. Use Tailscale, or give the profile a PIN and reach Ninaivu over HTTPS. More on remote access. p. 24

Someone forgot their password or PIN

Ninaivu stays at home and sends no email, so a new password comes from an administrator, not from a link.

- **A family member** presses **Forgot your password?** (or **Forgot your PIN?**) on the sign-in screen and then **Let an administrator know**. The console's Overview shows *Asked for help signing in*, and their card on **People** is marked. Press **Reset password** on that card: Ninaivu suggests a temporary password, which you tell them. They sign in with it and choose their own straight away. For a PIN, choose **PIN** under *Entry* and give them a new one.
- **An administrator** can be reset by another administrator the same way. The only administrator, or every one of them, sets a new password on the computer Ninaivu runs on: open the **Ninaivu Control Panel**, press **Reset a password…**, pick the name and type the new password twice. Or run `ninaivu reset-password NAME` in a terminal there. Either one signs that person out everywhere. Phones backing up with a phone key carry on; use **Sign out everywhere** on People if a phone was lost.

The browser warns about the certificate

Only when HTTPS is on (the console's Server page). Ninaivu then serves HTTPS with its own certificate authority. Trust it once per device: the tray's **Trust the HTTPS certificate**, or the Server page's button, and the tray page for doing it by hand on Windows and macOS. Or put Tailscale or a reverse proxy in front, which bring their own. On plain HTTP, the default, there is no certificate to trust.

Nothing appears after choosing the folder

The library is indexed in the background; a large one takes a while. The Overview's **Library** numbers and the Server page's log show it going. If the folder is on a drive that is not plugged in, the gallery says so rather than showing nothing.

The scan is slow

System → Performance says what this computer can do and what would help. On a small machine, turn off the passes you do not need on **AI models** (text in photographs and video moments are the expensive ones), or choose *power-saving* on the Server page while the household is using the computer. A scan can be stopped and picks up where it left off.

An import says some files could not be copied

Open **Library → Import** and the **Errors** list: each file says why, in plain words. The usual ones:

- *not allowed to read or write this file*: on a Mac the file may be locked (Get Info → Locked), or Ninaivu may need Full Disk Access (System Settings → Privacy & Security); then press **Retry failed**. Clips a camera or dashcam protected on its card are copied like any other, and the card is left alone.
- *the drive reported a read or write error*: the card, disk or cable may be failing; copy what you can from it soon.
- *the name or folder path is too long*, *may not accept this file name* or *too large for the archive drive*: the archive drive cannot hold that file (a FAT32 drive stops at 4 GB); a drive formatted APFS, NTFS or ext4 can.
- *the archive drive is full*: the import stopped and kept what it copied. Make room and press Start.

A file that failed for a passing reason was already tried a second time. **Retry failed**, or the next Start, tries the rest again.

Uploads say there is not enough free space

Uploads and phone backups are refused while the server's disk has less than 2 GB free, so the index and the thumbnails always have room. Free some space, or move the library to a larger drive.

A model would not download

The models come from GitHub and Hugging Face. Check the machine has internet at the moment of the download; the AI models page says which file failed. Once a model is here nothing is asked of the internet again.

Mugil says the copy did not restore

The weekly test brings a few files back and compares them. If it fails, **Mugil → Test restores** says which file and why; the usual causes are a changed Google password (sign in again on Mugil) and a full Drive.

Mugil says nothing is uploading

Encryption is on by default, and nothing is uploaded until the encryption key exists. Make it on the Mugil page and keep the recovery file.

Ninaivu will not start

The log in the Control Panel (or the tray's **View the log**) has the reason. The common ones: the port is taken by another program (Ninaivu then moves to the next free one and says so), the state folder is on a drive that is not mounted, or another Ninaivu is already running from the same state folder.

Straight after an update: the first start of a new version copies the settings and the index into `backups/before-update` in the state folder, and does not start if that copy cannot be made, usually because the disk is full. An older version started over an index a newer one changed does not start either, and says which copy to restore. The technical administrator guide has the steps.

Moving to another computer

System → Move to another computer lists the three things that make an installation and, on the new machine, points the index at wherever the library landed without re-indexing. The details.

Getting help

Ask whoever looks after your Ninaivu; the technical administrator guide says where to report a problem. Say which version (**System → Home & extensions** shows it under *This home*), what you did, and what the log said.

THE GUIDE · PART ONE

The family app

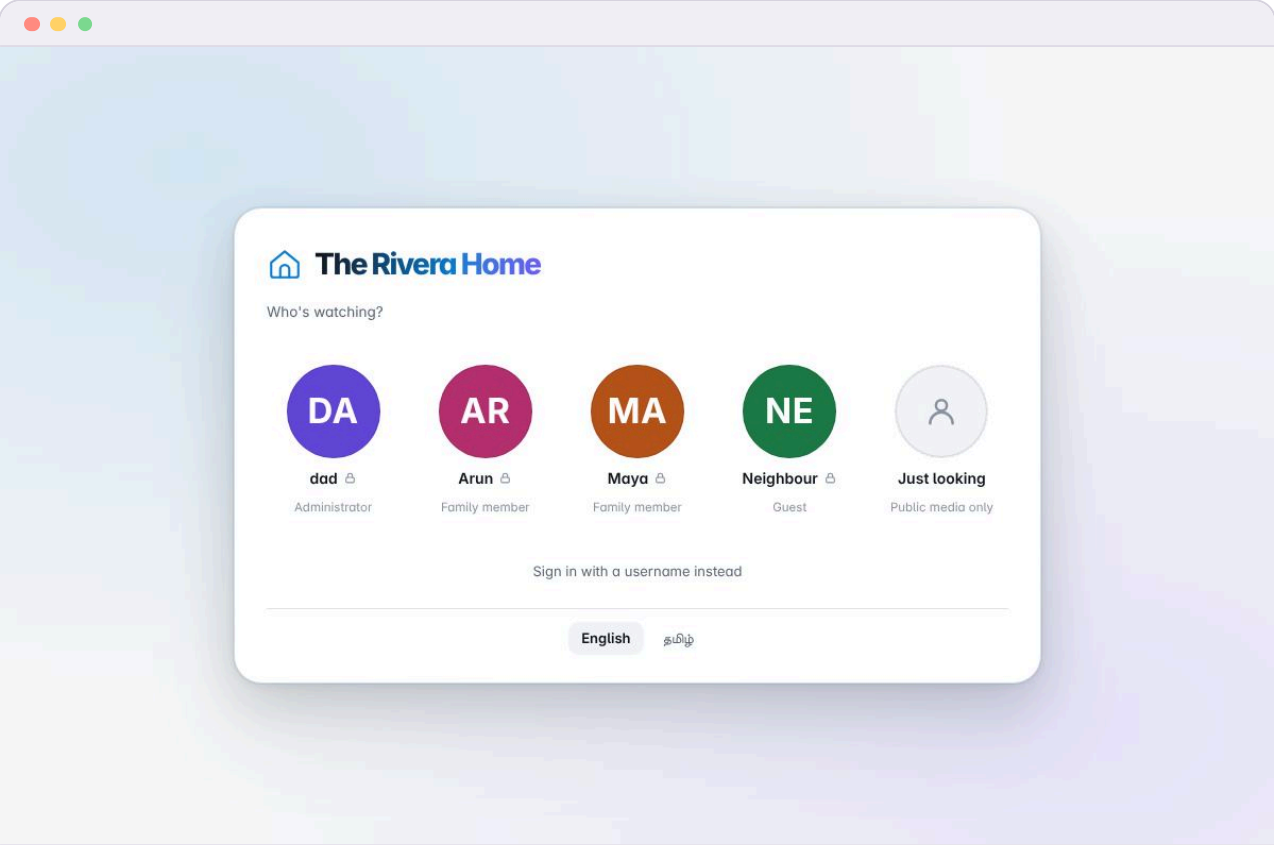
A

Everything a family member or guest needs, and nothing else. You do not need to know anything about servers to use Ninaivu: it opens like a website, and on a phone or tablet you add it to the home screen once and it behaves like an app.

Two things explain most of how it behaves:

- **Everything has a visibility.** Every photograph is *public*, *family* or *admins only*. What you see depends on which profile you came in as.
- **The management screens are somewhere else.** The administrator runs Ninaivu from a separate console on its own address. A family member or guest can change nothing about the library from the family app; an administrator signed in there can also delete and change visibility.

Coming in



The profile picker

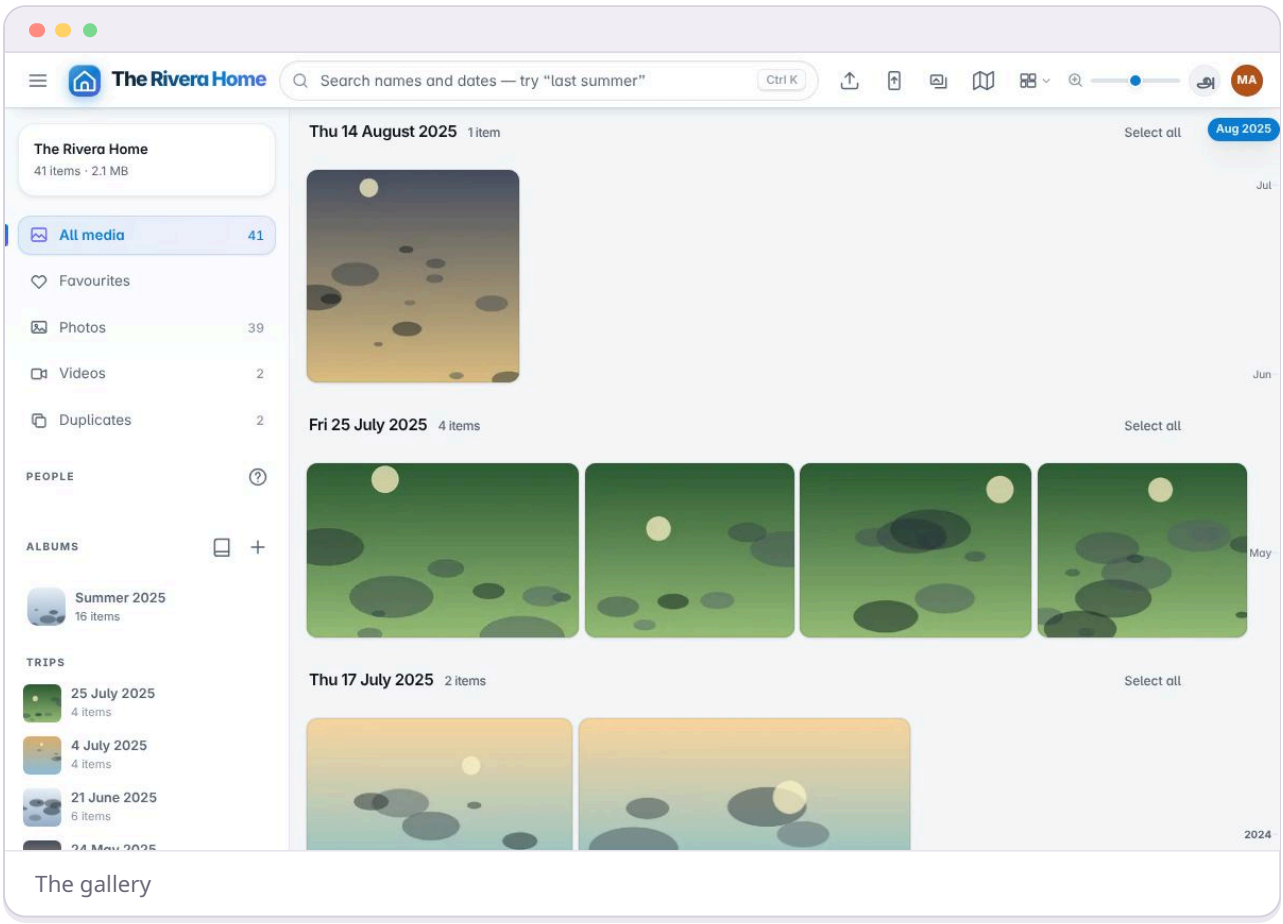
The family app opens on the household's profiles. Tap yours. What that costs depends on how the administrator set it up:

Tap to enter	nothing at all — right for the family tablet or a shared television
PIN	four to eight digits, shown as a small lock on your tile
Password	the full form, for guests and anyone who wants it
Just looking	with open browsing on, a tile that needs no profile and shows only what is public

The administrator has a tile here too, with their picture and a lock: tapping it asks for their password, never a PIN, and they still sign in to the console with their username. You can change your own name, colour and picture whatever your role.

Forgot your password or PIN? Press **Forgot your password?** under the box, then **Let an administrator know**. The administrator sees it in the console and gives you a temporary password, and you choose your own when you next sign in. Ninaivu sends no email, so this is the way back in.

The gallery



The gallery

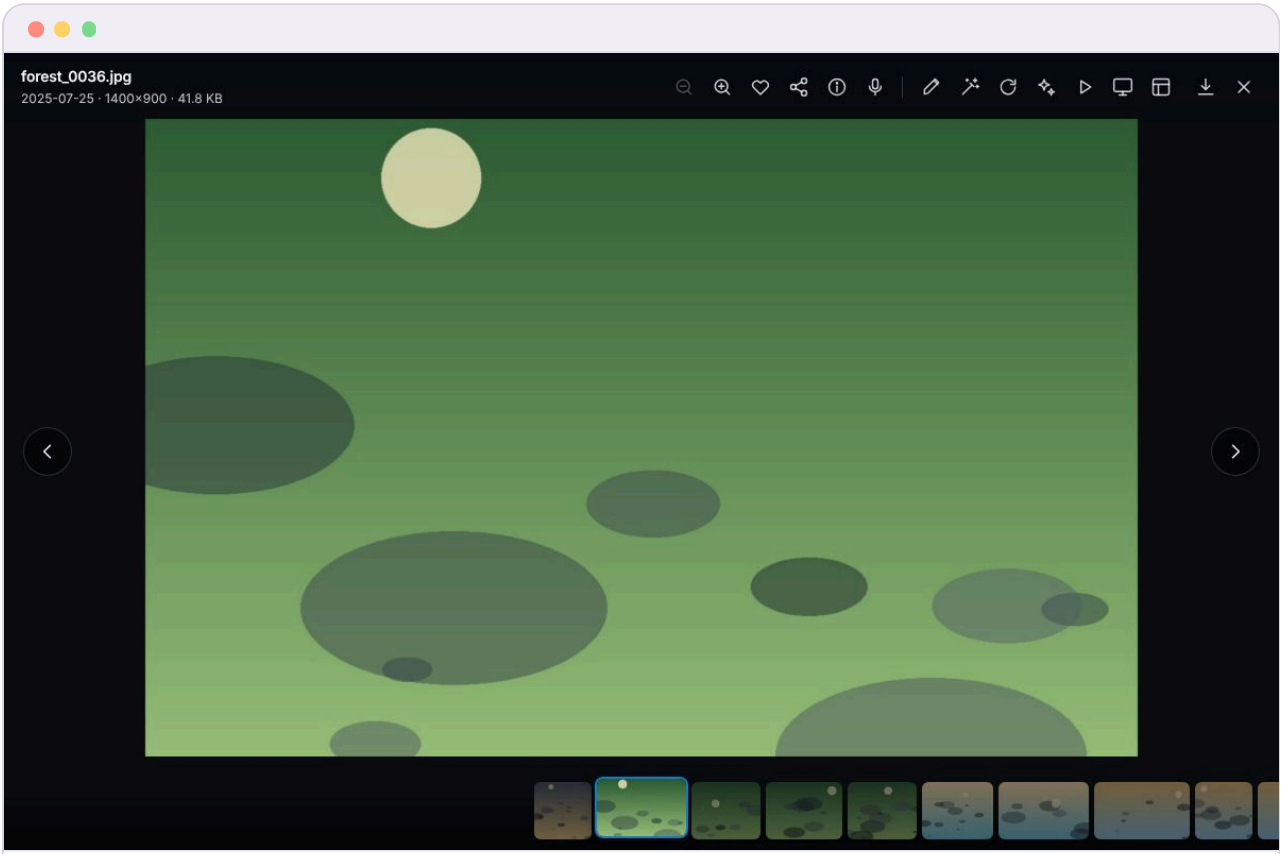
Everything you may see, newest first, grouped by day. Four layouts, and they are a real preference: **justified** suits mixed shapes, **masonry** tall photographs, **grid** scanning, **filmstrip** one long roll. `+` and `−` change the thumbnail size; `T` switches the theme.

The left column is the library's shape: photos, videos, favourites, duplicates the index found, albums, trips the dates suggest, then years and folders. Click any of them and it becomes a removable filter chip; chips stack, so *beach* + *2019* + a person is one click each. The strip down the right edge jumps anywhere in the library by month, as smoothly on a hundred thousand photographs as on a hundred.

Search takes plain words. On a Full machine (AI p. 26) the image model ranks results by how well the picture matches — "the beach at sunset" works as written, nobody tagged anything. On a Basic machine search covers names, dates, places, people and the words in file and folder names. The map button shows every located photograph grouped by place.

Selecting. Click the circle on a tile, Shift-click for a range, `X` on the focused tile, or ⌘/Ctrl+A for everything shown. With a selection you can favourite, download or add to an album at once; an administrator also changes visibility in bulk.

A photograph



The viewer

The whole photograph fitted to your screen, with its date, place and people, and the actions your role may take.

- **Zoom** with the wheel or a pinch, drag to pan, double-click to zoom to a point; `0` puts it back. Arrow keys or a swipe move on.
- `I` — **details**: camera, lens, ISO, aperture, where the date came from, the location if the file carried one.

- `S` — **find similar**: up to twelve photographs that look like this one — the other attempts at the same moment. It needs search by description (AI p. 26); without it the panel says the photograph is not analysed yet.
- `F` — **favourite**. Favourites are private to you, even when several people share the same folder. Guests cannot favourite.
- `R` — **rotate** (Shift for the other way). A turn saved by a family member is saved for the whole household; the file itself is not touched unless an administrator writes the turn into it.
- `D` — download the original (family members and administrators).
- **Share link** — a link for this photograph, with an optional password, that shows exactly what you could already see.
- **Edit photo** — the Photo Studio, below (family members and administrators). The button beside it opens **Sudar**, the AI studio, below that.
- `Space` — a slideshow. `K` — the **Ambient Frame**: full screen, a slow drift across each picture, a clock in the corner, a new photograph every few seconds. It is the reason to keep an old tablet on the kitchen wall. `Esc` comes back.

A **Live Photo** from an iPhone is one item with a Live badge. It plays once when you open it; press and hold the photo (or the **LIVE** button) to play it again with its sound, and let go to go back to the photograph. On a computer, resting the mouse on a Live tile in the gallery plays it there. Ninaivu pairs the photo and its clip while indexing, even when they were filed in different day folders, and a clip a browser cannot open is converted for it.

A camera set to **RAW+JPEG** takes two files of every shot. They are one item in the gallery, showing the JPEG, with a RAW badge; **Download RAW** under **More** in the viewer takes the RAW file. Hiding or deleting the photograph does the same to its RAW.

Voice stories

The microphone button on the viewer's bar (`V`) opens the **stories** told about a photograph or video — Paati saying "this is my father's shop in 1968", in Tamil or English — with how many there are on the button.

- **Record a story** asks for the microphone then, and only then, and records up to five minutes; **Stop**, listen back, **Record again** if you like, and **Save story**. **Upload audio** takes a recording already on the device instead (up to 25 MB).
- **What is said** is optional: type the words, in any language, and the gallery's search finds the photograph by them. Nothing turns the voice into text by itself, and no recording or word of one is sent to any outside service.
- **Who is speaking** starts as your own name; change it to whoever is talking.

- A story is seen by exactly whoever may open its photograph: a photograph kept for administrators keeps its stories to them too. Guests can listen but not record. Whoever told a story, or an administrator, can delete it.
- A share link of the photograph carries its stories, to listen to.
- Browsers allow recording only over a secure (`https`) address; over a plain `http` address, use **Upload audio**, or ask the administrator to turn on HTTPS. Recordings are kept as the browser made them, so one made on an older iPhone may not play in another browser, and the other way round.

Press `?` anywhere for the full list of keys.

People

If faces are on, a row of people appears in the sidebar. Click one to see that person's photographs. The count is *yours* — how many you are allowed to open — so two people in the house may see different numbers for the same person. Naming people and confirming matches are the administrator's job on the console; in the family app, People is read-only — with one exception, below.

Who is this? Ask the family

The question-mark button beside **People** (family members and administrators) opens **Ask the family**. Choose up to 20 faces nobody has named and press **Make the link**; send the link by WhatsApp or e-mail to whoever might know — Paati does not need an account. The page it opens shows only those faces, in English or Tamil, with a box for a name and a note ("my uncle Raman, at Chennai, around 1975"). It never shows the names the family already uses, and it shows the whole photograph only if you ticked **Show the whole photograph too** — never one only administrators may see. A link lasts 30 days and can be withdrawn under **Links sent**; a photograph hidden after the link was made drops out of it.

Answers wait under **Answers**, saying who answered and when. **Yes, it is …** joins the face to somebody already named, **Accept as a new person** makes a new one, **Dismiss** throws the answer away. Accepting confirms the face just as the console's review does, so Ninaivu can then suggest that person across the whole library. A family member reviews the answers to their own links; an administrator sees all of them, and can also ask from **Faces** on the console.

Family tree

The tree button beside **People** opens the family tree: a row for each generation, a line from each child to their parents and between couples. Choose a person at the top to add or remove their parents, children or spouse; nobody can be made their own ancestor. Tap anyone in the tree to see

their photographs. The tree holds only the people you can see in your People row, so somebody who appears only in photographs you cannot open is not in it.

Albums

Any family member can make an album and add to it. Albums hold references, not copies: deleting an album never touches a photograph. An album can be shared as a link, with an optional password.

Photo books

Open an album, a trip or a person and choose **Make a book** to have a print-ready PDF made: a wedding, Pongal, Deepavali, a sixtieth birthday, a holiday. Choose a template (Plain, Wedding, Pongal, Deepavali, Birthday or Holiday, each with its own colour and ornament), a size (A4 portrait or landscape, or a 20 or 30 cm square, all at 300 dpi), the title, how many photographs (12 to 120, 36 to start with) and whether each has a caption with its date and who is in it. **Add 3 mm bleed** if the print shop asks for it.

Ninaivu then picks the photographs: blurred ones, screenshots and repeats are left out, faces and favourites come first, and the picks are spread across the whole occasion in the order they were taken. Untick any you would rather leave out, then **Build the book**. It is made on this computer, a page at a time, and nothing is sent anywhere. Only photographs you can see yourself go in, and only the names of people you can find under People.

Finished books are kept under **Books** (the book button beside Albums) to download again or delete. A book is yours: nobody else in the family sees it in their list. Tamil titles need a Tamil font on the computer Ninaivu runs on; if there is none, the dialog says so and Tamil words are left out of the PDF.

Sending photographs

Family members can upload from the gallery; the files wait for the administrator under **Review → Uploads** before they join the library, and a name that already exists gets a short suffix rather than overwriting anything. On a phone added to the home screen, the backup screen sends the photographs and videos you choose: keep the page open while it sends — a web page cannot read the phone's library by itself or keep going in the background — and it skips what is already here and resumes where it stopped. The limit is 512 MB per upload. Guests cannot upload.

To have the phone back up without opening anything, make a **phone key** for it and give it to a sync app such as PhotoSync: Backing up by itself p. 21.

Scanning old prints

Scan old prints (in the top bar, or under **More** on a phone) turns printed photographs into photos in the library. Take a photo of one print, or of several laid out on a table with a little space between them, or choose photos you already took or scanned — up to ten at a time. Ninaivu finds each print, flattens it if the photo was taken at an angle, turns it the right way up and straightens it; if it cannot tell the prints apart, it uses the whole photo.

Untick anything that is not a print, then choose:

- **Fix faded colours** (on by default) brings back the blacks and whites and removes the red or yellow cast old prints get.
- **Restore faces** and **Make it bigger** use the same AI models as Sudar's Enhance tools; they are greyed out until an administrator downloads them under **AI models**.
- **Keep the original photo** keeps your phone's photo too, in `Scanned prints/originals`.

Then say when the prints are from: a year (`1975`), a decade (`1970s`) or a date (`1975-06-12`), or leave it empty. If the picture-search model is installed, Ninaivu suggests a decade — it is a guess from how the prints look, so check it. The prints are saved as new photos in `Scanned prints/<year>` (or `Scanned prints/Unknown year`) and are dated as you said; nothing already in the library is replaced. A family member's prints wait for the administrator under **Review → Uploads**, like any upload. Nothing is sent outside the house. Guests cannot scan prints.

The Photo Studio

Edit photo opens the Photo Studio: light, colour and detail, on this device, with the original kept. Every slider means the same thing as the slider of that name in Sudar, because the two share one engine.

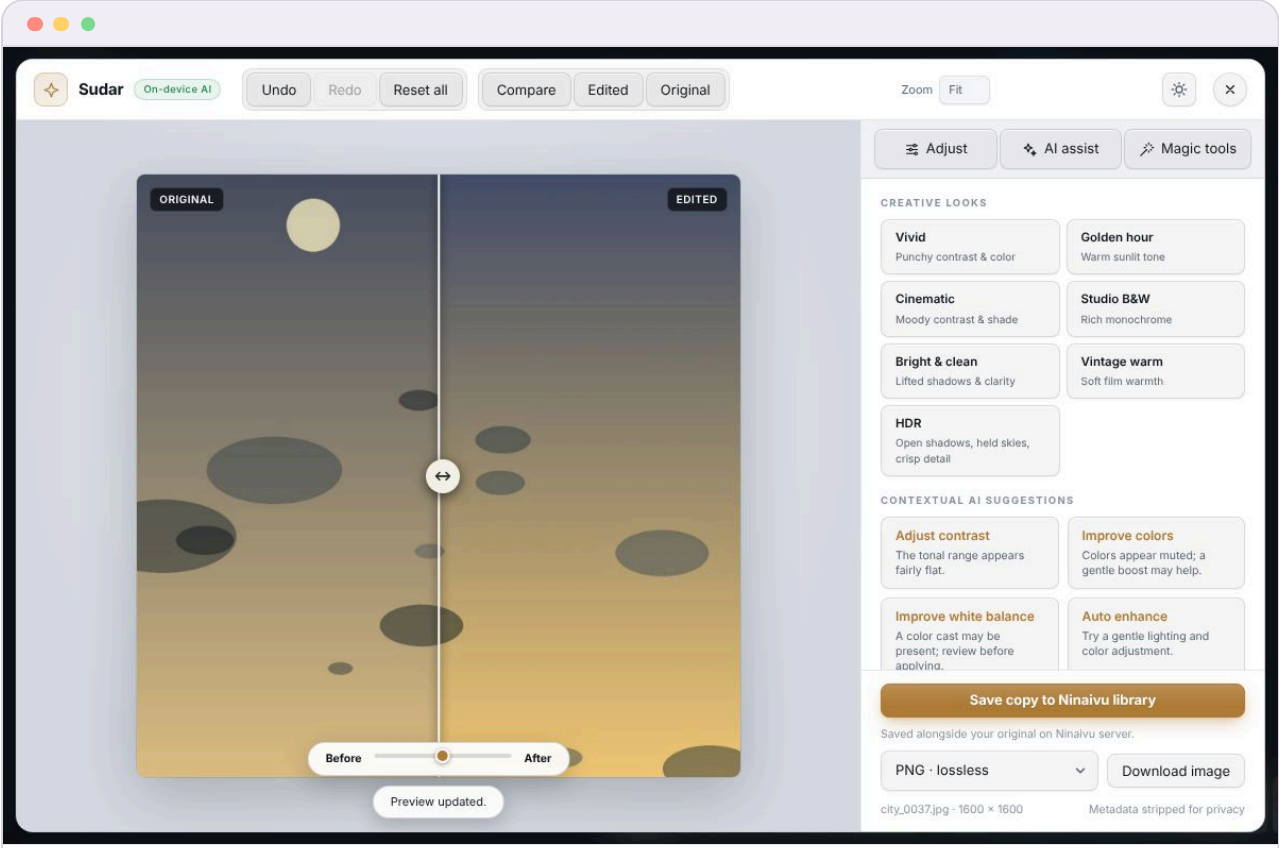
- **Light** — a histogram at the top; exposure (40 on the slider is one stop), contrast, highlights and shadows judged by their neighbourhood so a face in front of a bright window opens up without a pale ring round it, whites, blacks, dehaze; and a **tone curve** drawn over the histogram, for all channels or one, with points added by clicking and removed by dragging them off the square.
- **Colour** — white balance and tint; **vibrance**, which knows where brown skin of every complexion sits and leaves it nearly alone, and saturation; an eight-band **colour mixer** (hue, saturation, luminance); **black and white**, whose tones come from the mixer's luminance

sliders; **split toning**; and **Natural skin tone**, which judges the colour of the whole photograph by the skin in it and moves warmth and tint only as far as skin of any complexion looks natural.

- **Detail** — clarity (negative softens), sharpening on lightness only with radius and masking, luminance and colour noise reduction, film grain, and a vignette with midpoint and feather.
- **Skin** and **Hair** — the people in the photograph, each against their own skin: below p. 44.
- **Looks** — ten for family photographs, each shown as a small picture of the photograph being edited and laid over your own sliders with an amount: Natural, Festival, Golden hour, Soft portrait, Backlit rescue, Revive old print, Film, Monsoon, Classic black and white and Warm black and white. None raises the lightness of skin or drains its colour.
- **Crop** — straighten, ratios, flip across and flip upside down. **Export** — full size, or 3840, 2048 or 1080 pixels on the long edge.

The photograph is worked a strip at a time, so a 24-megapixel file fits in the browser's memory. Saving writes a copy in the same folder with what the camera wrote — make, model, lens, exposure settings, place — carried over; the original is never changed. An administrator's copy joins the library at once; a family member's waits under **Review → Uploads** for approval.

Sudar, the photo studio



Sudar

Sudar (சுடர், *glow*) edits a photograph in the browser: looks, suggestions measured from the picture, light and colour sliders (exposure, contrast, shadows, highlights, dehaze; vibrance, saturation, white balance; clarity, sharpness, noise, vignette), straightening and crops (square, 16:9, 4:5, and 9:16 for a story), with the original on the left and the edit on the right. "Make it warmer" in plain words works too. The original file is never changed; a copy can be saved beside it, and a family member's copy waits for the administrator's approval like an upload.

Any result can be saved. A cut-out, a blurred background, a colour pop (the subject in colour, the rest in black and white), an upscaled, restored or colourised picture, a generated preview: each has **Save to library** beside **Download**, and goes the same way as the sliders' edit — published beside the original for an administrator, into the approval queue for a family member. **Object removal** and **Clothing colour** put their result back on the photograph with **Apply to photo**, so the sliders and Save are still there.

Enhance tools — **Upscale**, **Restore faces**, **Colourise** — are always on the card; one that is not set up is greyed out, and the line under it says which model it needs and that an administrator adds it (AI p. 28). With both of the last two, **Revive old photo** restores the faces and then colourises an old print in one tap.

With the Creative Studio extension on (AI p. 26), Sudar also does generative edits — including a new **hairstyle**, which is drawn rather than adjusted, from ten ideas under the prompt, and **Add hair**, which paints where hair should be and has the AI server draw it — object removal and upscaling, on this computer or on an AI server at home. Nothing leaves the house.

The Creative Studio (the Magic tool of that name, not the extension) makes things from photographs, in the browser: a restoration story and a then-and-now comparison, a family story album with a recorded memory, a postcard, a collage in three layouts, a cinematic slideshow with music, selective colour, a sheet of named versions, a recipe card and a photo calendar. Photographs come from your files or **from the library**; the picture creations come as a landscape, a square or a 9:16 story, and can be **saved to the library** like an edit, or downloaded, or printed. The album, slideshow and comparisons export as a web page that works on its own.

Skin and hair

Open **Skin** or **Hair** in the Photo Studio, or **Skin & hair retouch** in Sudar — they are one set of tools — and Ninaivu looks for every face in the photograph, on this computer, with the face model the People page already uses. Across the top are the people it found: **Everyone**, then each face. Choose one and a slider moves for that person alone (a dot beside it says so); choose Everyone and it moves for everybody who has not been given a setting of their own. **Improve faces** gives each person what stood out about them — a face in shadow is lifted, a yellow or pink cast is turned back towards skin, a shiny forehead is calmed, grey hair is covered — and nothing else. It is one step of Undo, and every slider it moved is still a slider.

Look younger, at the top of the Skin panel, does in one tap what these tools can honestly do about age, for whoever is chosen and by the amount you set: smoother and more even skin, calmer shine, brighter under-eyes, grey covered, strands brought out, thin hair filled. It only moves the sliders below, so each can be changed afterwards; it lightens nothing and redraws nothing, and the person stays exactly who they are. Ninaivu has no age model, on purpose: the ones that exist redraw the face.

What these tools do and do not do:

- **Each person is judged against their own skin.** A fair face and a very dark one in the same photograph are not pushed towards the same colour.
- **A bindi, kumkum, sindoor, sacred ash or sandal paste is left exactly as it is.** Nothing here smooths, recolours or blurs it, and its colour cannot spread into the skin beside it.
- **Nothing makes skin lighter or fairer.** *Face light* is exposure for a face that is in shadow; skin of every colour is lifted by the same amount, so a dark face stays dark. It is only offered when the whites of the person's eyes show the face really is dim — never because the skin is dark.
- **Eyes, brows, lips and teeth are not touched.** *Soften* takes blemishes and leaves pores; a fold, an eyelid or a mole is kept.

- **Hair and beard:** *Strands & shine*, *Fuller hair* (fills thin hair: where light shows through between strands, the gaps are brought down to the strands' tone with their texture kept; it adds no hair where there is none, and never reaches skin), *Cover grey*, and *Hair colour* (black, dark brown, brown, chestnut, henna, burgundy or any colour). When hair cannot be told from what is behind it — black hair against a black wall — Ninaivu asks the background-removal model, if it is installed (AI p. 26), where the background is; without it, it says so rather than guess, and you paint the hair in with **Add** and **Remove**, in the Photo Studio and in Sudar alike. What you paint goes with the picture if you crop or turn it.
- **Add hair.** The third brush in the Hair panel: paint where there should be hair — a receding hairline, a thin crown — and it is drawn from the hair beside it, the same strands, colour and shine running on; *Added hair* says how much shows. It is this person's own hair, made on this computer, and it works best for a band of a finger's width or two along the hair that is there. For more than that — a bald patch, a whole new head of hair — Sudar's **Add hair · generative** Magic tool paints the area and has the AI server draw it, when the Creative Studio extension has a *Paint and ask* workflow.
- **Natural skin tone**, in the Photo Studio's **Colour** panel, judges the colour of the whole photograph by the skin in it — indoor light is yellow, a shaded courtyard is blue — and moves the warmth and tint sliders only as far as the edge of what skin of any complexion looks like.

Finding faces needs the face model (**AI models → Finding and grouping faces**, part of the essential set). Without it the panel says so, and the brush still works.

Language

The language button switches between English and Tamil (the sign-in and lock screens have it too). It is saved on your profile, so every device you use follows it.

The screen lock

After 15 minutes without use, the app locks; your PIN or password opens it again, and anything in progress carries on meanwhile.

Sound recordings

Sound files in the library — voice notes, voicemail, dictation swept up from old drives — are administrator-only, always. A photograph is looked at on purpose; a recording plays out loud the moment somebody taps it, in a room with other people in it.

On a phone

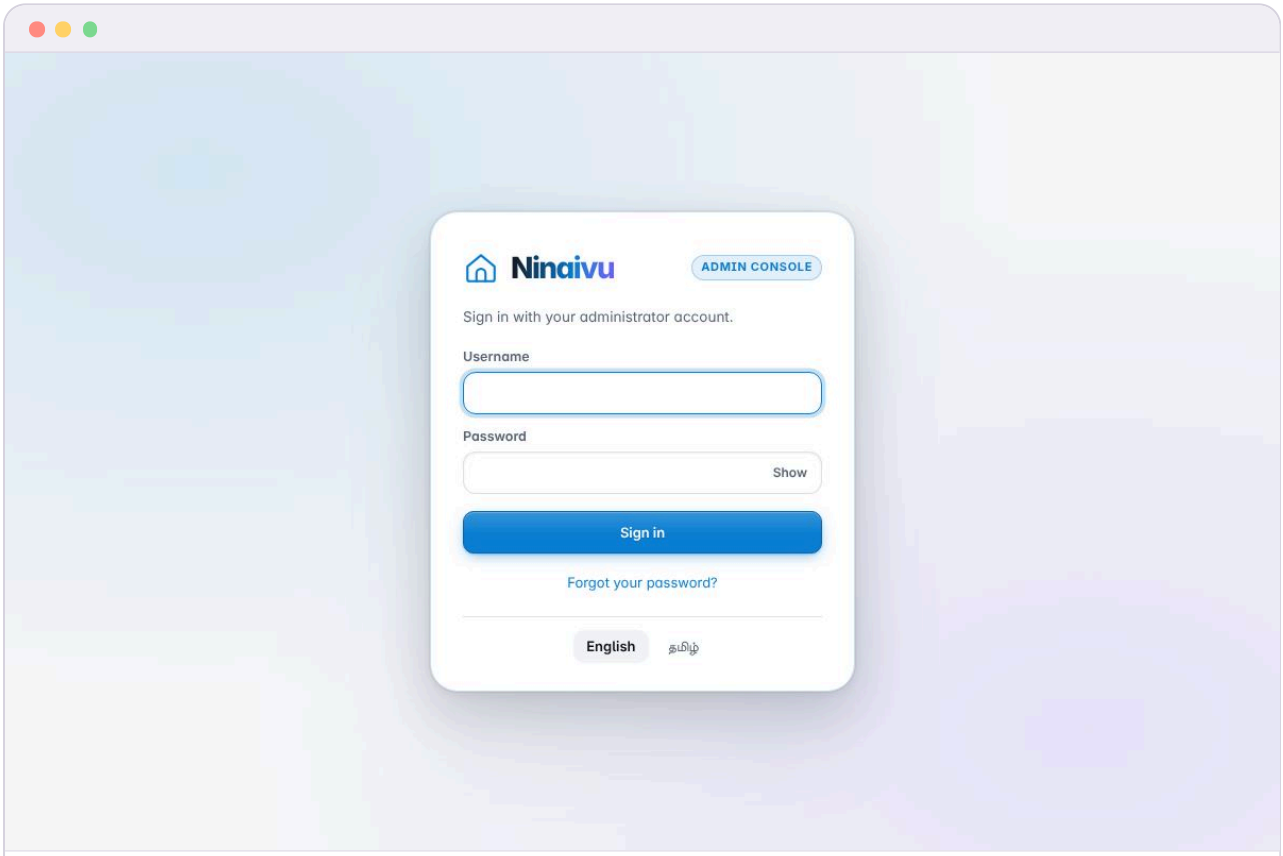
Open the address the administrator gives you while on the home Wi-Fi, then **Add to Home Screen**. It opens like an app and works offline for what it has already shown. Away from home, it works only if the household set up remote access p. 23.

THE GUIDE · PART TWO

The console

B

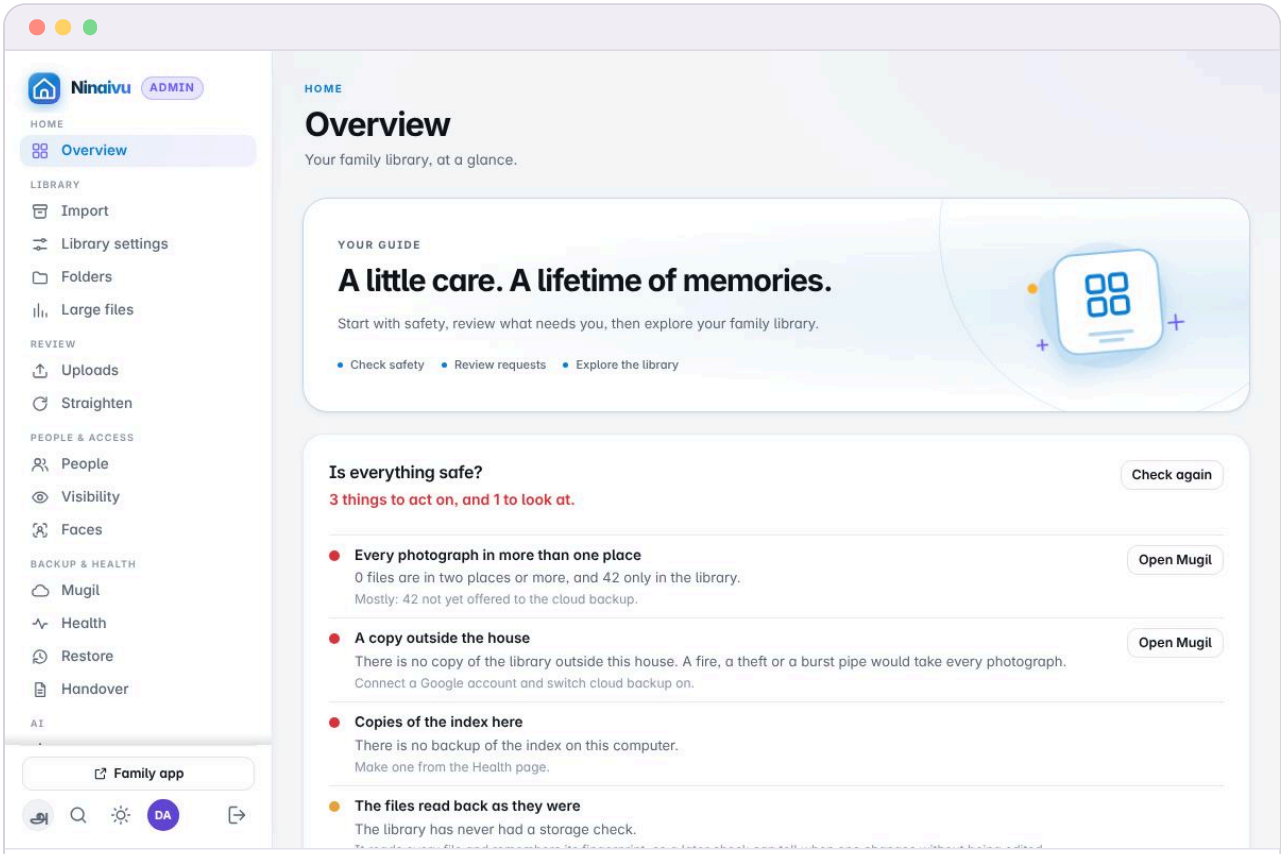
Everything the administrator does: getting a library indexed, deciding who sees what, and the tools — Import, Faces, Mugil, Health. The console answers on its own port, on the computer Ninaivu runs on only until you open it to the home network on the Server page, and always takes a password, never a PIN. When the server cannot be reached, it shows *Ninaivu is Offline* rather than the browser's own error; it keeps nothing, so it never works offline.



Signing in to the console

> **Two apps, two ports**
>
> The family app and the console are two applications in one process. The console's routes are *absent* from the family app, not merely forbidden: people, visibility rules, Import, Mugil and settings exist only on the console port. An administrator signed in to the family app can still delete photographs and change their visibility from the gallery. A session made on the family app cannot open the console, even an administrator's.

Home — Overview

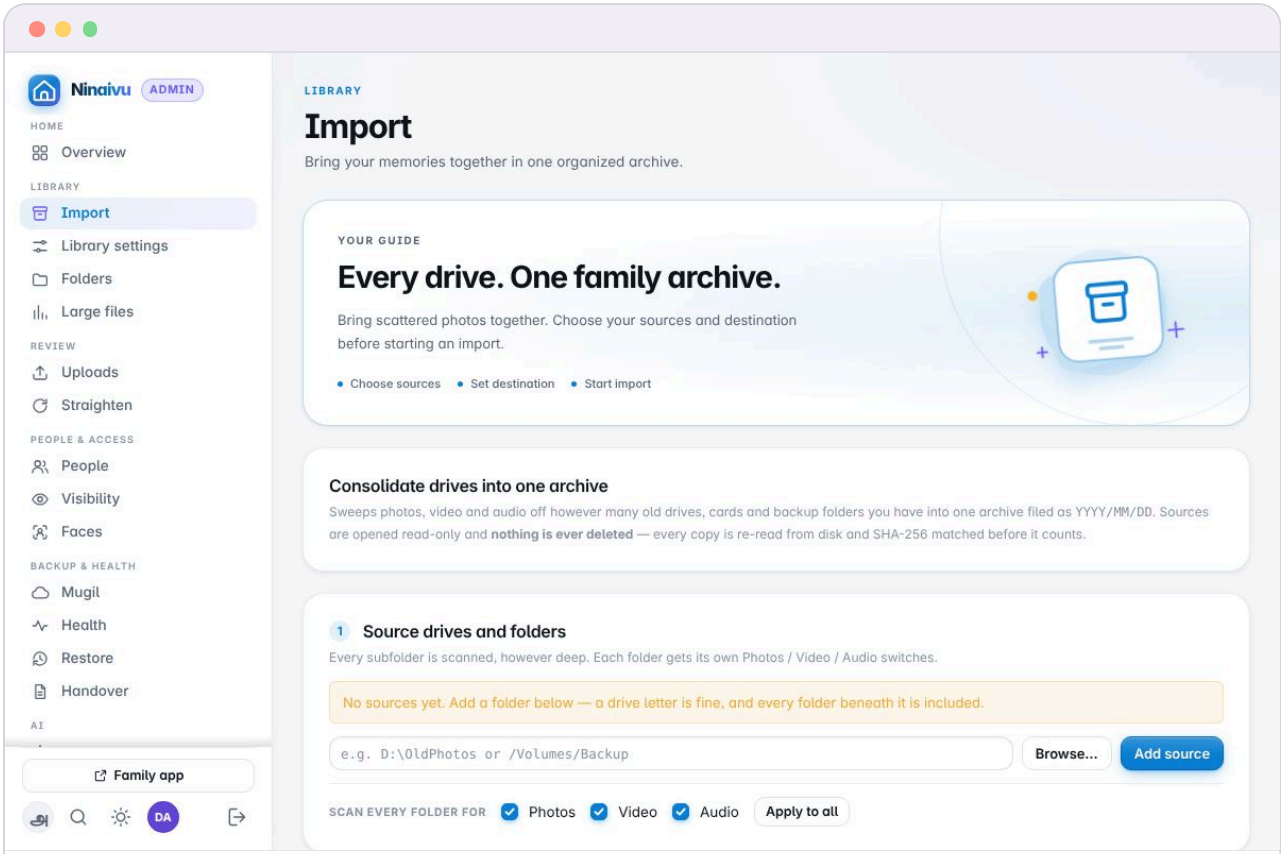


Overview

The first screen answers two questions before anything else. **Is everything safe?** runs the household's safety checks — a copy outside the house, a backup that restores, copies of the index, healthy drives — and says what to do about each. **Needs you** counts what is waiting for a person: uploads to approve, faces without a name, photographs that may be on their side, things that went wrong. Then the numbers, and what each role would actually see. The Overview is also where the first day p. 12 opens, once.

Library

Import



Import

The first Library page on purpose: a library usually begins with old drives, memory cards and backup folders, swept into one archive laid out as `YYYY/MM/DD`. Every file is hashed on the way in and read back to verify the copy; duplicates are recognised by content and left in place, logged; sources are only ever read. **Dry run** decides everything and writes nothing. **Start consolidation** also resumes: an interrupted run picks up where it stopped. Files under 60 KB — icons, thumbnails — are left alone.

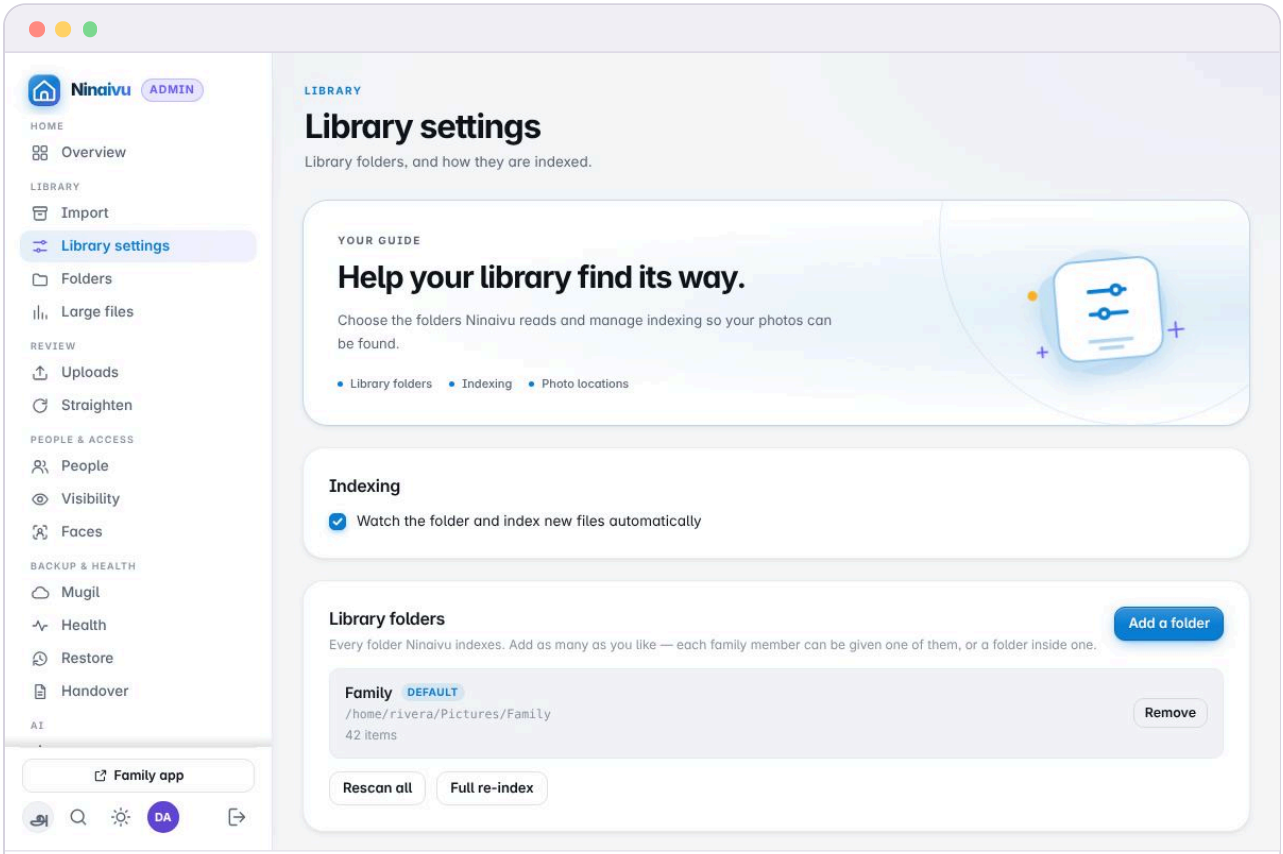
An import cut short by a restart of Ninaivu or the computer carries on by itself as soon as Ninaivu is back; the page and the activity strip say *resuming after restart*, with how many files were already done, and the count shows the files found so far. Refreshing the console stays on the page that was open, so the numbers are not lost.

A file that fails for a passing reason (still being written by a sync app, open in another program, a drive slow to answer) is tried once more, 30 seconds after the rest. Whatever still fails is in the **Errors** list, with the reason in plain words first: the file was locked or Ninaivu needs Full Disk Access, the file was gone, the drive reported a read error, the name is too long for the archive drive,

the file is larger than a FAT32 drive allows. **Retry failed**, or the next Start, tries them again. Clips a camera or dashcam protected on its card (locked, on a Mac) are copied like any other file; the copy is not locked and the card is left as it was. When the archive drive fills up, the import stops with *the archive drive is full*, keeps what was copied and verified, and carries on with Start once there is room.

When the run has finished the page offers to **add the archive to the library**. A Google Photos Takeout export comes across whole: its sidecars give the dates, places and descriptions, and once the archive is indexed the page offers to **make its albums** from the copies it kept. The **Archive Guardian** re-reads a rotating sample of archived files every day and says when one has changed.

Library settings



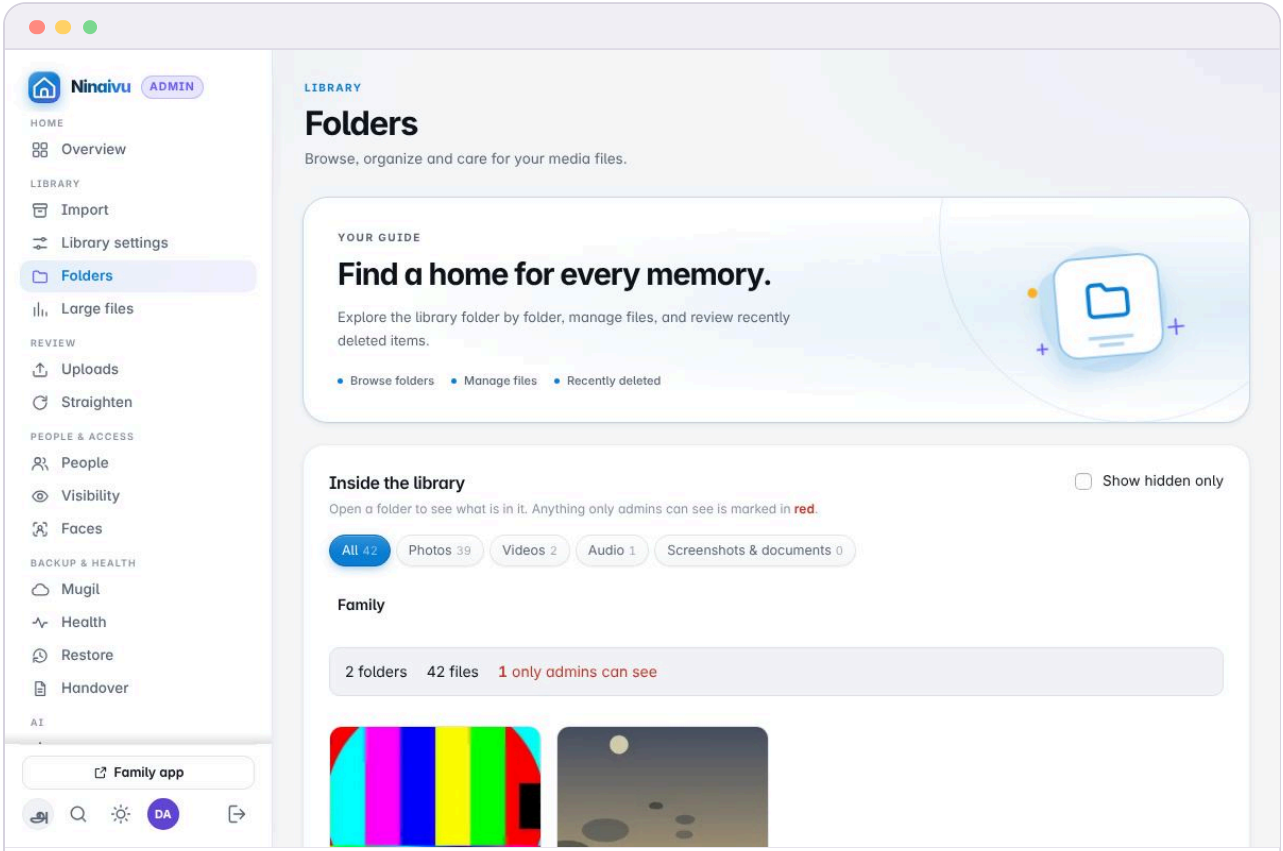
Library settings

Which folders make up the library, and how they are indexed: whether Ninaivu watches them for changes, whether hidden files are indexed, what the scan works out by itself. Add as many folders as you like; a folder on a drive that is unplugged stays a library folder and the gallery says so rather than showing nothing.

Where the dates come from: the camera's own date first, then the container's metadata, then a Takeout sidecar, then a date in the file name, then a `YYYY/MM/DD` folder path, and only then the file's own creation or modification time. The viewer's details panel says which one was used.

When a library folder cannot be written — an NTFS drive on a Mac, say — edited copies, approved uploads and phone backups go to `~/Pictures/Ninaivu` instead, which joins the library on first use.

Folders



Folders

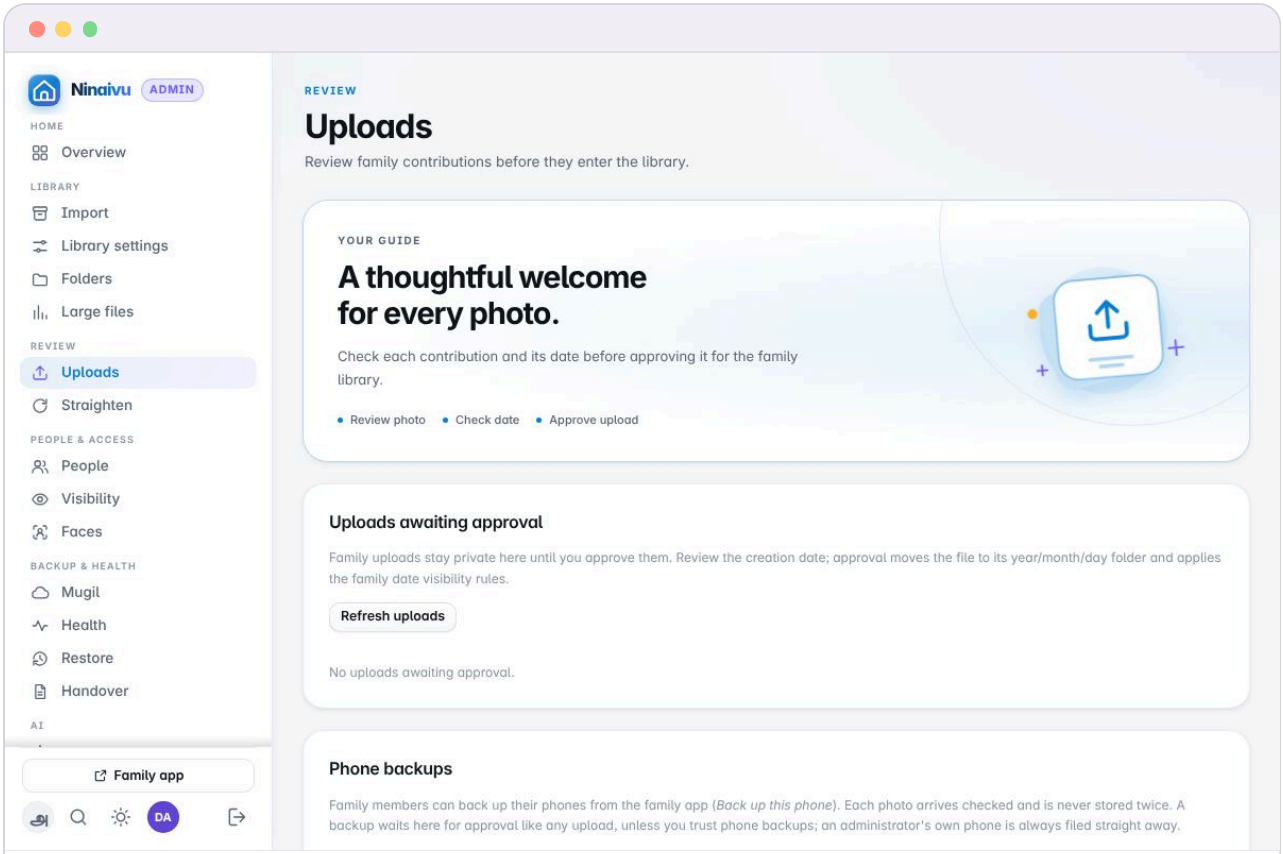
The library as folders, with what is in each and each folder's visibility, and the **recycle bin**. Deleting in Ninaivu moves a file into a `_deleted` folder beside the library, where it can be put back with its faces and albums intact. Nothing is erased for good unless you empty the bin here (it asks first) or set a number of days on Advanced settings.

Large files

The files taking the most space, to look over once and mark as reviewed. Tick the files you want (or *Choose all*) and use the buttons above the list. For videos, **Compress** (recommended) saves a smaller MP4 copy beside the original and leaves the original as it is, and **Replace** which puts the smaller copy in the original's place after checking it plays, keeping the original in `_deleted/_originals`; Replace asks for your password. Both need ffmpeg and are greyed out without it.

Review

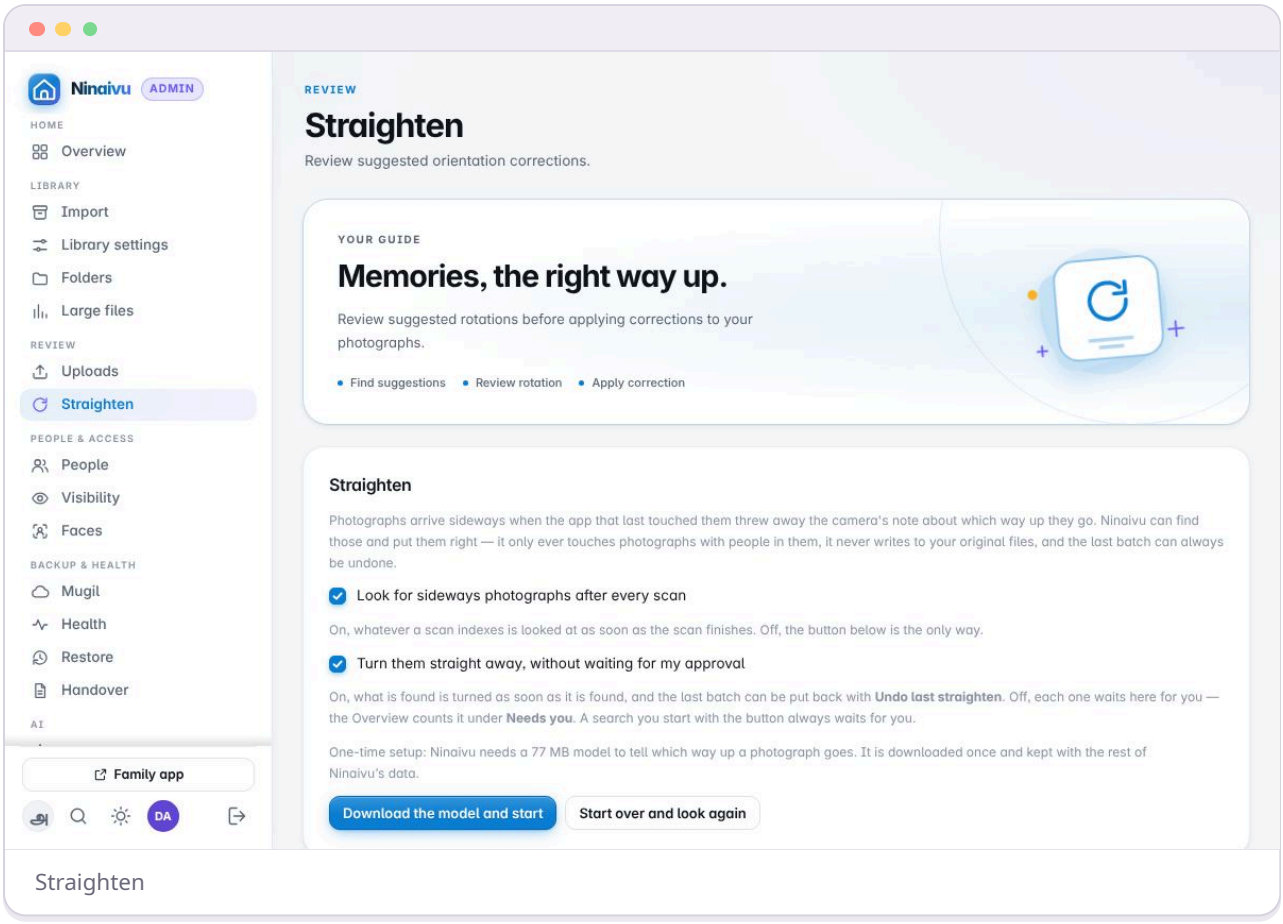
Uploads



Uploads

Photographs the family sent from their phones, and edits saved in Sudar, waiting to be filed or refused; and the phones that have sent backups. Nothing a family member sends joins the library without passing here, unless you tick **Trust phone backups**; an administrator's own phone skips the review.

Straighten

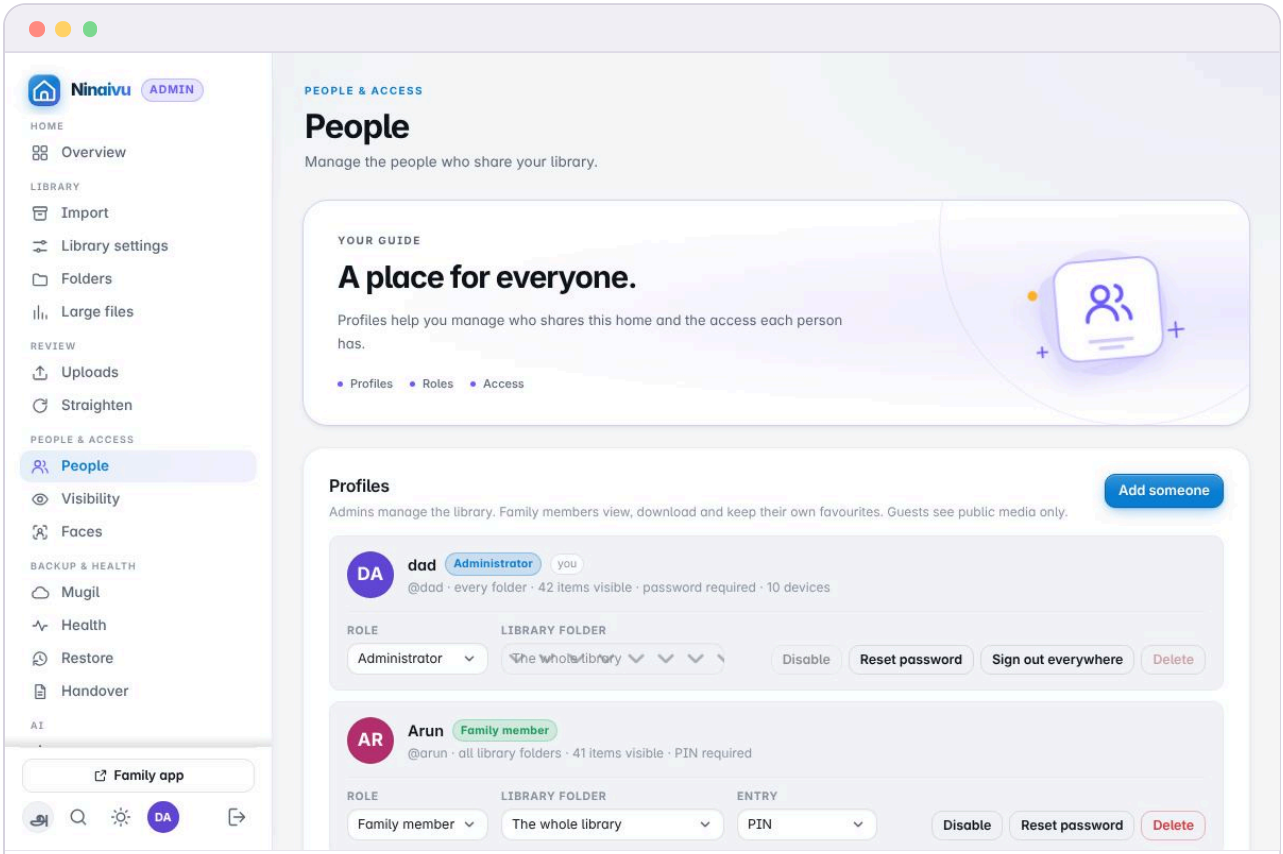


Straighten

Photographs the model thinks are on their side, strongest guess first, each shown the way it would be left. Click one to skip it; **Straighten them** turns the rest in one batch. It only ever suggests a quarter turn, and by default only for photographs with a person in them — a found face is a second witness that this way up is the way a person stands. By default the looking happens by itself after every scan, over what the scan indexed, and what it finds is turned straight away, as one batch. **Undo** puts back the last batch exactly, thumbnails included. Two switches at the top change this: one turns the automatic looking off, leaving the button; the other keeps what it finds here for approval instead of turning it. A search started with the button always waits for you.

People & access

People



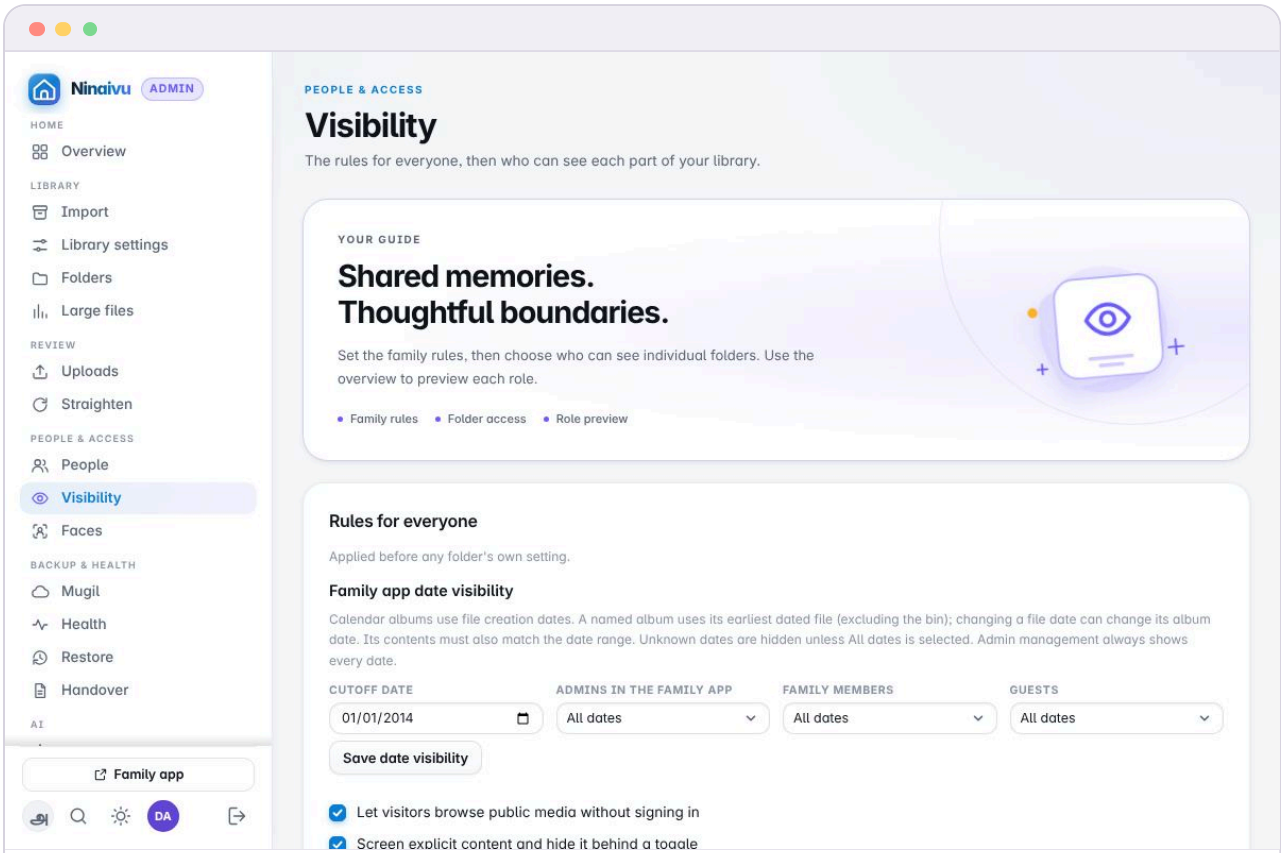
People

The household's profiles: one administrator, family members, guests. Each has a role, a way in (a PIN, a password, or a tap), and optionally a folder of their own — any library folder, or any folder inside one. Assignment and visibility stack, and the narrower always wins: a guest given a folder still sees only what is public in it.

Disable is reversible and keeps the person's favourites; **delete** is final and removes the profile, its favourites and every open session. Neither touches a photograph. You cannot delete yourself or the last administrator. Disabling a profile or resetting its password ends its sessions everywhere at once.

Somebody who pressed **Forgot your password?** on the sign-in screen is marked *asked for help signing in*, here and on the Overview, until they have a new password or PIN. **Reset password** gives a temporary one to tell them; they choose their own at their next sign-in. A forgotten administrator password, with no other administrator to reset it, is set from the Control Panel's **Reset a password…** on the computer Ninaivu runs on. The steps. p. 30

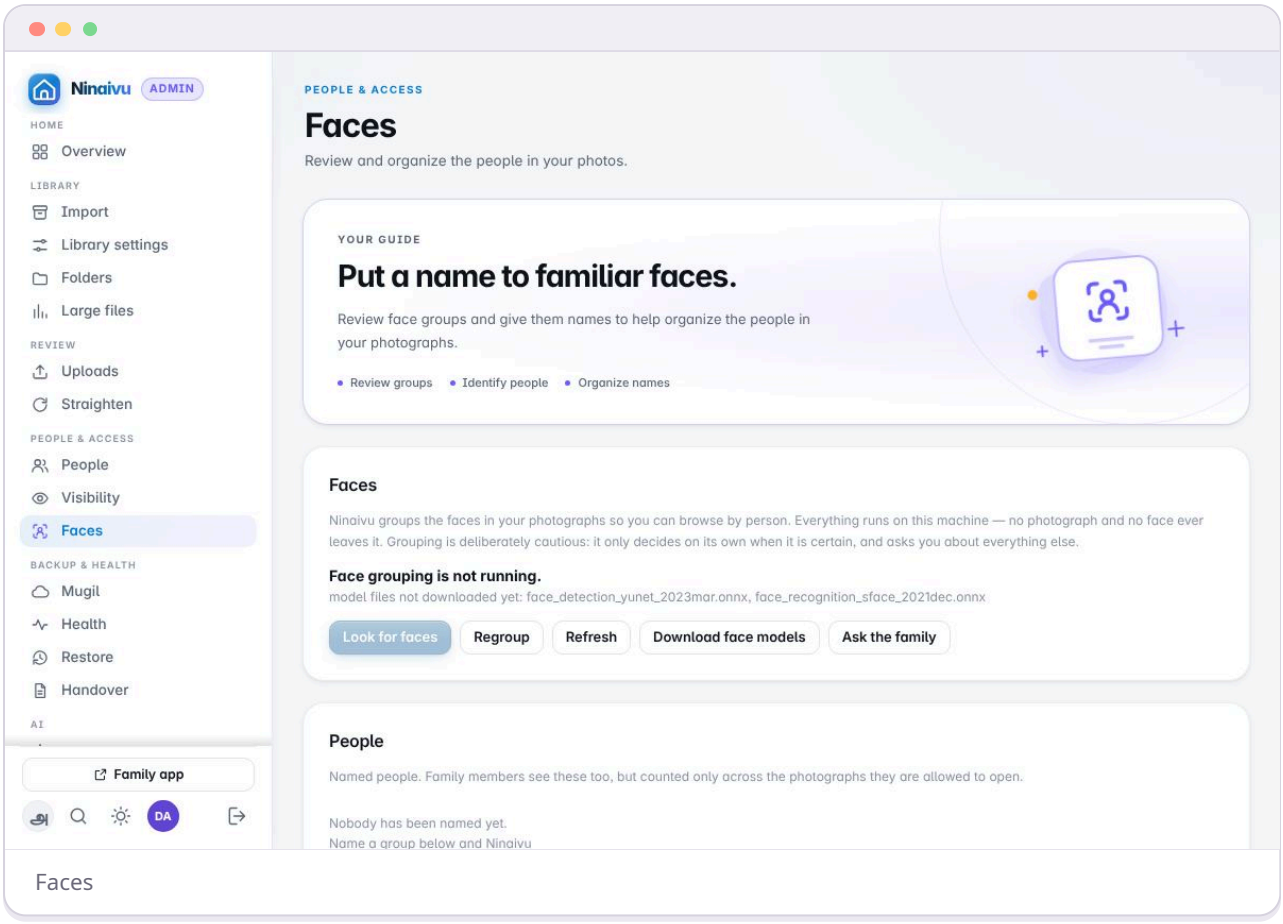
Visibility



Visibility

Who sees what. The rules for everyone come first — whether visitors may browse without signing in, whether explicit content is screened, how far back in time each role may look — and then each folder's own setting: everyone, family, or admins only. New media is *family* until somebody says otherwise, so nothing reaches a guest by accident. A folder's setting is remembered as a rule, so files scanned into it later inherit it. Setting a folder rewrites everything beneath it, so the previous state is written down first and every change can be undone in one step.

Faces

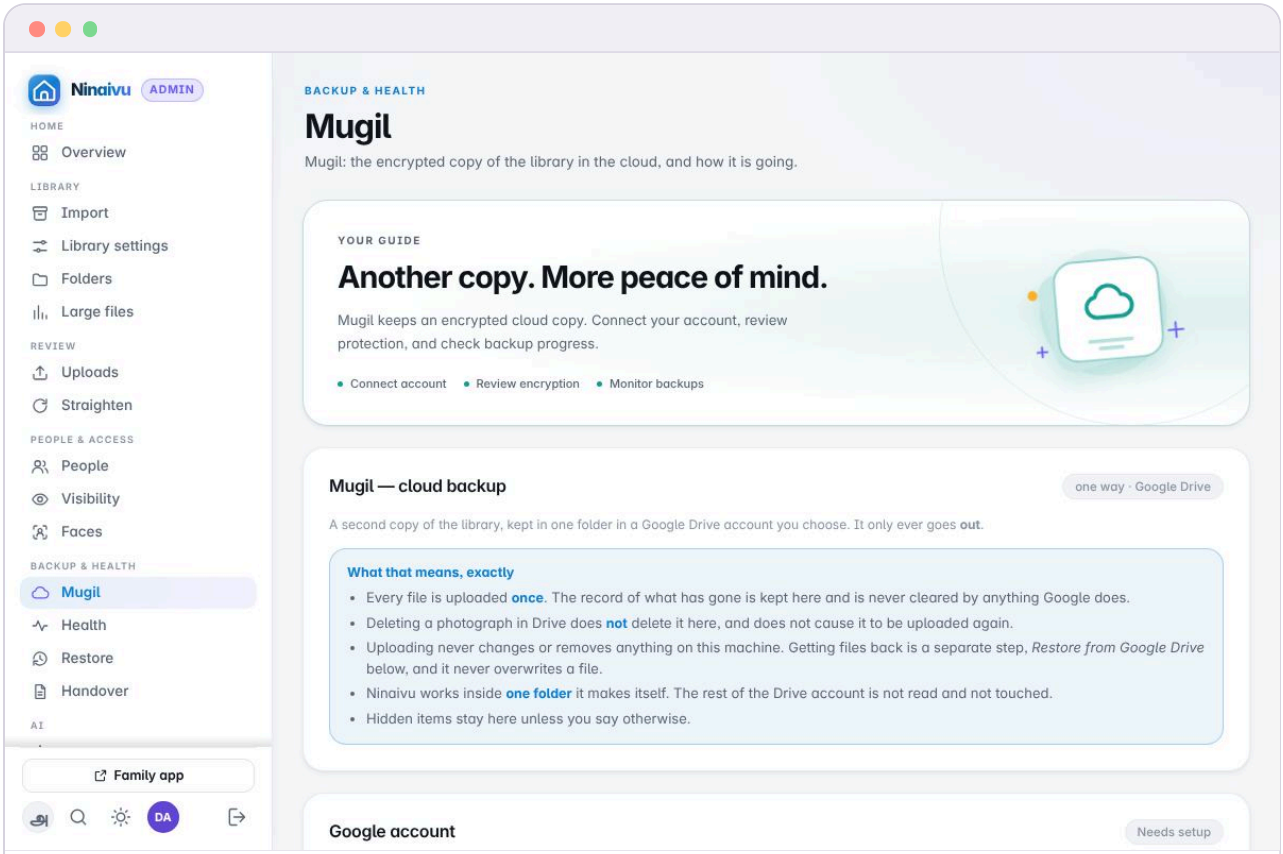


Faces

The people the face matcher found, grouped. Name one face and the rest of its group follow; groups waiting for a name are listed for a quick pass. Faces smaller than 50 pixels are skipped on purpose, which is the trade that keeps the ones it shows reliable. Faces are found and matched on this computer; the names travel only inside Mugil's encrypted copy of the index.

Backup & health

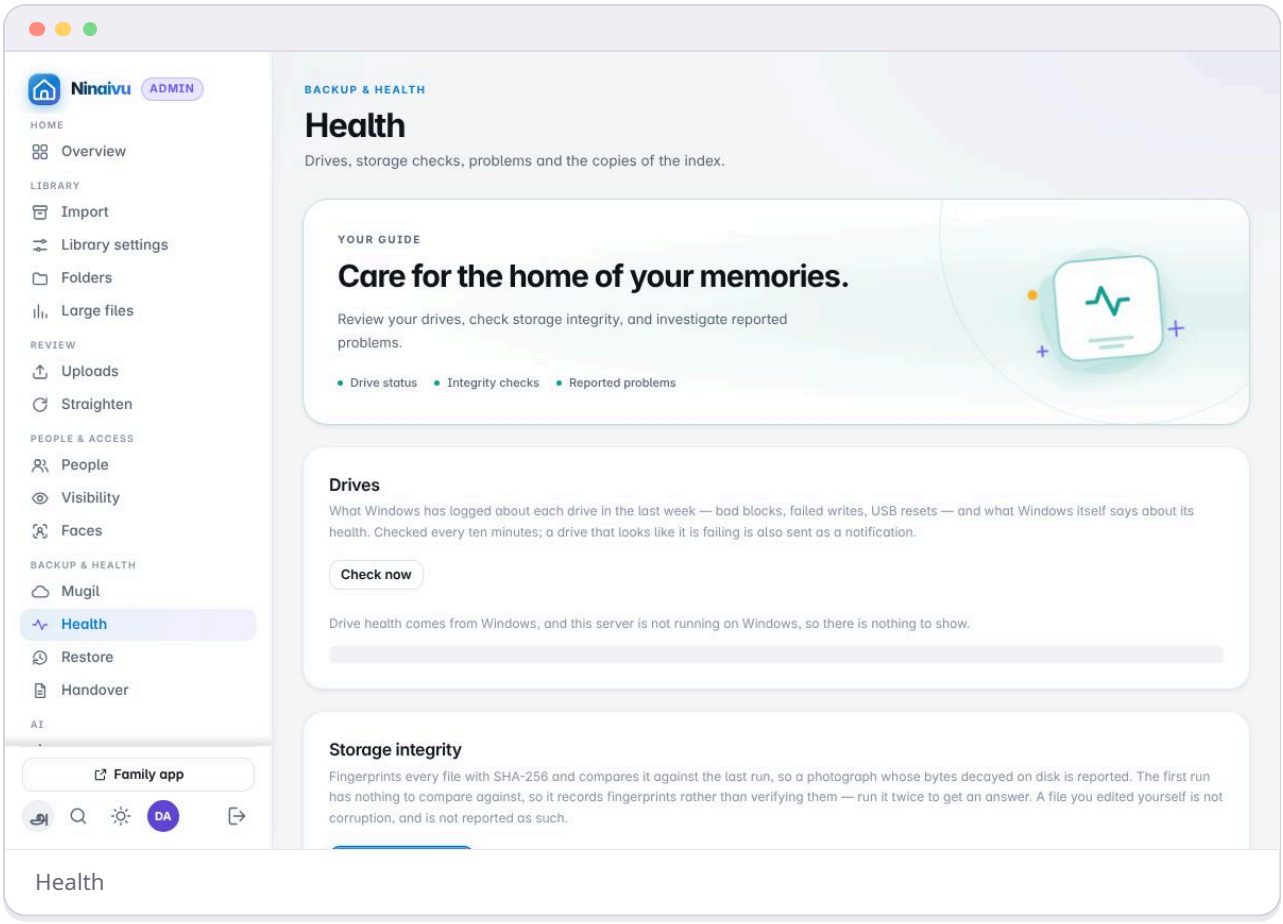
Mugil



Mugil

Mugil (முகில், *cloud*): the encrypted copy of the library on Google Drive — the account, the encryption key and its recovery file (nothing is uploaded until the key is made), what is sent and what is kept back, speed and hours, and **Test restores**: the weekly test that the copy restores, with its history. More on backup. p. 18

Health



Health

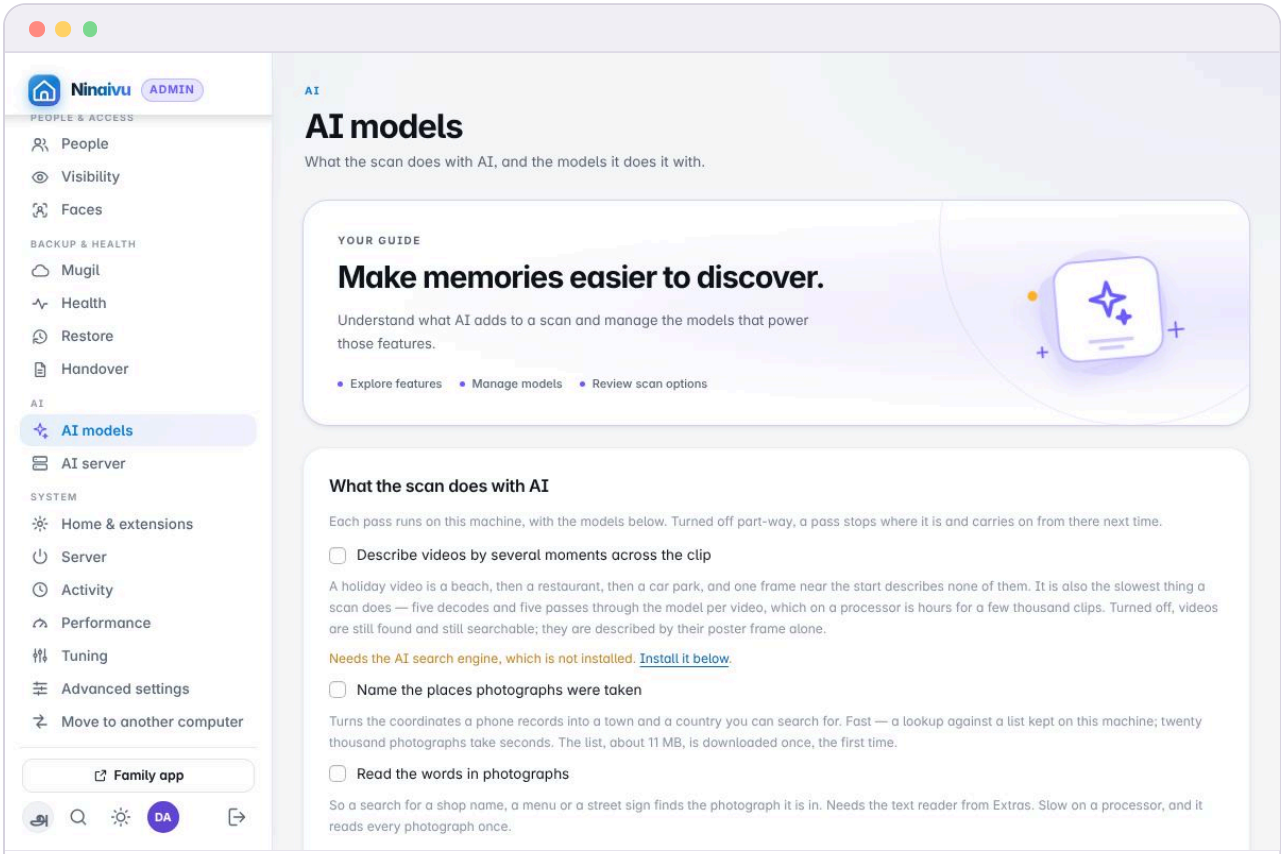
The drives and their SMART state, the storage check that reads every file back against its fingerprint, anything that went wrong since the last start, the copies of the index kept on this computer, and how the household is told when something needs them — by email or a webhook. The storage check, which takes hours on a large library, steps aside while new photographs are indexed and then carries on. **A photograph, once a week:** on the morning you choose, Ninaivu picks the best photograph from that day in a past year and emails it to the family; a day with nothing worth sending stays quiet.

Restore

Bringing photographs back from the cloud copy, in three steps: what to bring back, where to put it, and the encryption key. Nothing here ever overwrites a file that is already there.

AI

AI models



AI models

What the scan does with AI — naming places, reading text, finding faces, describing videos by several moments — each switch saying which model it needs and whether that model is here, then the models themselves with their downloads. Everything on this page runs on this machine. More on AI. p. 26

AI server

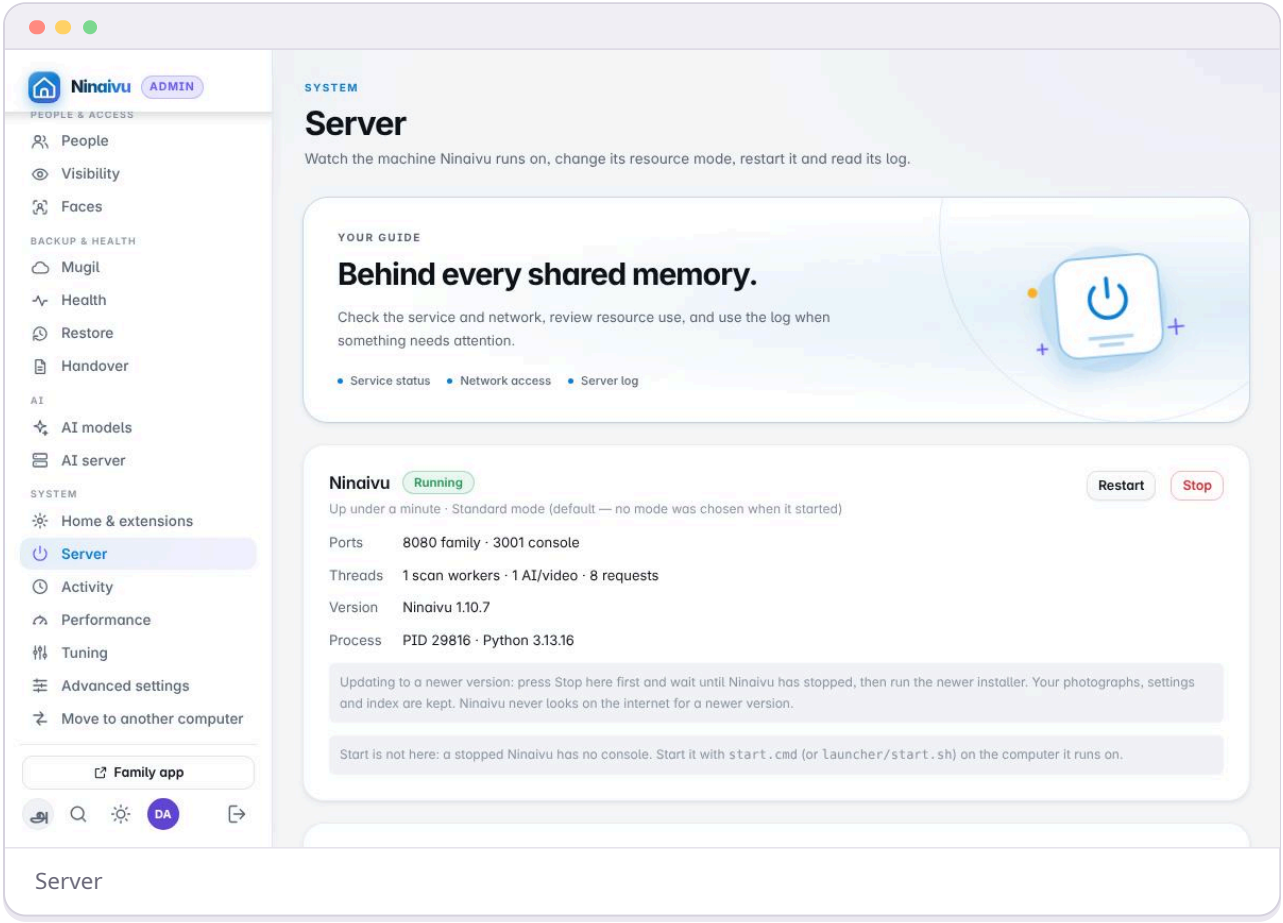
For a household with a second, stronger computer running ComfyUI: hand the heavy jobs to it. This page belongs to the Creative Studio extension and appears only while it is on.

System

Home & extensions

This home's name and the version of Ninaivu; the **extensions** installed, each saying on its switch what it does when it is on, and whether family members may send photographs to one that goes outside the house; and **Extras**, the optional packages — search by description among them.

Server

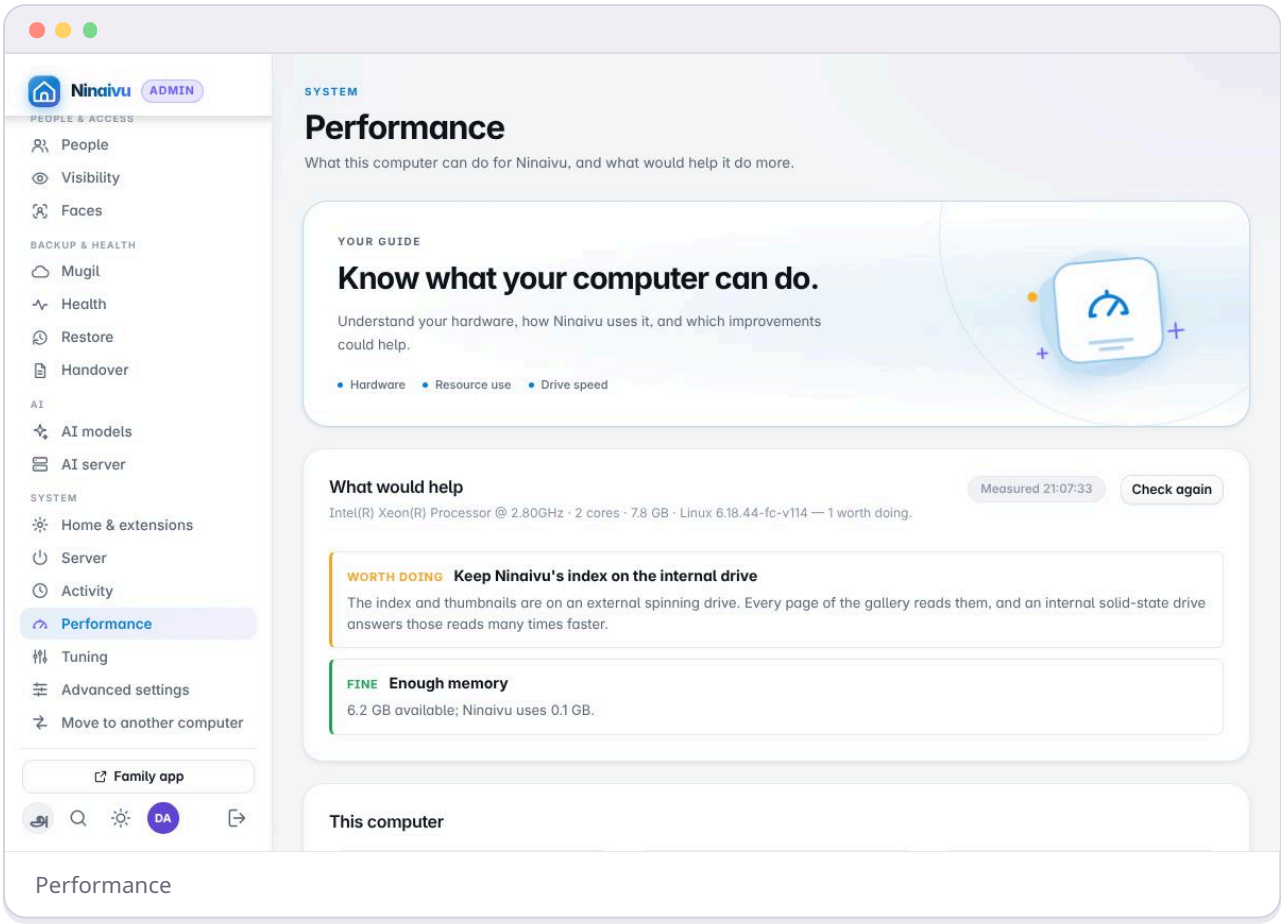


Whether Ninaivu is running and where it answers, its version (press **Stop** here before installing a newer one), the addresses to give the household, **Away from home** — how the household reaches Ninaivu from outside (remote access p. 23) — **Network access** and whether the console may be opened from other devices at home too (it never opens from the internet; see remote access p. 24), the resource mode (standard, performance, power-saving), a restart, the log, the **HTTPS** switch (off by default; turning it on restarts Ninaivu), and, while on HTTPS, the certificate. The ports are shown here; they are set when Ninaivu starts (`--port`, `--admin-port`).

Activity

What is running now and who is using the computer, so the backup and the scans can make way for the household. **Balanced** gives way to video playback; **quiet** waits whenever anyone is using the app; **overnight** keeps the heavy work for the night hours and runs it flat out then.

Performance



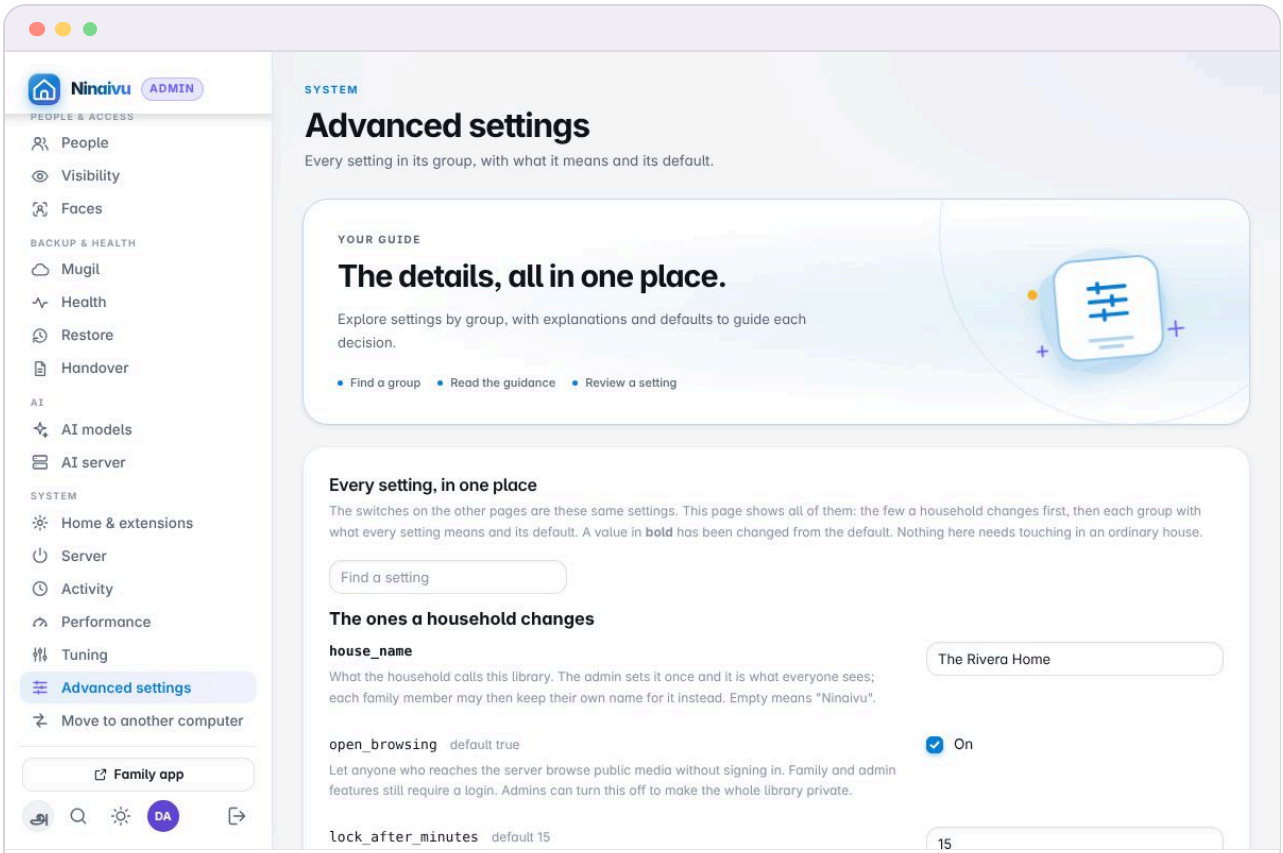
Performance

What this computer can do for Ninaivu — which kind of computer it is (Basic or Full p. 26), the processor, memory, graphics, drives, how fast the last scans went, what is waiting — and what would help it do more.

Tuning

How hard Ninaivu works this computer. At every start it measures the processor, memory, graphics and drives and picks a profile: **Small box** (a Raspberry Pi or a small computer), **Everyday computer** or **Powerful computer**. **Peak performance**, which uses every core and up to 95% of the memory, is only chosen by hand. The page shows what was measured, what the numbers are expected to use, and every number with where it came from; each can be changed, or put back to automatic. The details.

Advanced settings



Advanced settings

Every setting Ninaivu has, in six groups, with what each means and its default. The switches on the other pages are these same settings. Nothing here needs touching in an ordinary house.

Move to another computer

Moving Ninaivu, or telling it the library has moved, without re-indexing everything. The details.

The command line

Nothing in this guide needs a terminal: the installers, the Control Panel and the console do it all. The command-line options, the maintenance commands (a copy of the index, a restore, a new password for a forgotten administrator) and the environment variables are in the technical administrator guide, in English.



Ninaivu

Your family's photographs, at home.

THE GUIDE, ONLINE
javajaga-usa.github.io/Ninaivu/

DOWNLOAD
github.com/javajaga-usa/Ninaivu/releases

REPORT A PROBLEM
github.com/javajaga-usa/Ninaivu/issues

Ninaivu 1.10.9 · MIT licence